

Practical Experiments in **BIOLOGY** 2



Satya Naidu



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resources



Practical Experiments in

BIOLOGY

2



Satya Naidu



Acknowledgements

I wish to thank Carol Andrews for reviewing the original drafts and Pam Robinson for her suggestions and for providing materials for the experiments.

K. Satya Naidu

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INTRODUCTION

Practical Experiments in Biology Book 2 is part of a four-book series that presents experiments from different areas of study that are relevant to senior Biology students. Book 2 focuses on classification, dissection, digestion, reproduction and ecology.

Teacher’s notes and useful websites for each experiment are provided at the start of the book. These include strategies for enhancing student understanding of the experiment and suggestions for overcoming difficulties. Internet addresses for relevant websites are also provided to encourage students to undertake further research.

The experiments are presented in a self-contained format that Biology students will find easy to follow. Each experiment includes a summary of relevant background knowledge and is structured according to aim, hypothesis, equipment, safety rules, method, results, analysis of results and conclusion. Tables and space for diagrams are provided for students to record their results. The clear, step-by-step structure will encourage the efficient use of experimental time and give guidance to students on how to write scientific reports. Suggested answers and expected results have also been provided where appropriate.

CONTENTS

Teacher's notes and useful websites	4
1 Classification of animals	6
2 Classification of plants	10
3 Construction of a dichotomous key for arthropods	13
4 Dissection of a sheep’s pluck	16
5 Dissection of a rat	20
6 Digestive enzymes	25
7 Asexual reproduction in plants and animals	30
8 Food chains, food webs and food pyramids	34
9 Surveying an ecosystem	38
10 Suggested answers	42

TEACHER'S NOTES AND USEFUL WEBSITES

Experiment 1: Classification of animals

Suggestions for teachers

- Students should have a thorough understanding of the principles related to the classification of animals into vertebrates and invertebrates.
- Make the students aware of the characteristics that are used to classify animals
- Provide live specimens, preserved specimens, photos and reference books.
- Show a video on classification.
- Make students aware of appropriate websites on classification.

Websites

<http://web.ukonline.co.uk/webwise/spinneret/pot/invertebra/invert.htm>

<http://kidport.com/RefLib/Science/Animals/AnimalIndexV.htm>

Experiment 2: Classification of plants

Suggestions for teachers

- Students should have a sound background knowledge of the principles of the classification of plants and the characteristics of various divisions.
- Biology reference books, appropriate images and diagrams and live specimens should be made available.
- Students should be familiar with using stereomicroscopes.
- Show a video on plant classification.
- Make the students aware of appropriate websites about plants and the classification of plants.

Websites

<http://www.dscc.edu/bwilliams/Biology2/biology2labplantking.htm>

<http://home.manhattan.edu/~frances.cardillo/plants/intro/test.html>

<http://www.learn.co.uk/default.asp?WCI=Unit&WCU=2415>

http://www.swishweb.com/Plant_Kingdom/plant01.htm

Experiment 3: Construction of a dichotomous key for arthropods

Suggestions for teachers

- Students should have a thorough background knowledge of how to construct and use two types of dichotomous keys.
- Students should be familiar with the classification and characteristics of different classes of arthropods.

Websites

<http://health.allrefer.com/pictures-images/arthropods-basic-features.html>

<http://www.kendall-bioresearch.co.uk/class.htm#crustacea>

<http://members.aol.com/YESedu/arthrocl.html>

<http://web.ukonline.co.uk/webwise/spinneret/pot/invertebra/invert.htm>

<http://ourworld.compuserve.com/homepages/BMLSS/EdibleC.JPG>

Experiment 4: Dissection of a sheep's pluck

Suggestions for teachers

- The pluck consists of the two lungs, the heart and the liver. It can be ordered from a butcher. Make sure that it is intact and no cuts have been made to the lungs or the heart.
- Students should have a thorough understanding of the structure and functions of the human respiratory, circulatory and digestive systems before the dissection.
- Students should have prior experience in dissecting organs and know how to use dissecting instruments.
- Resources such as biology text books, charts and models should be made available for comparison.
- Make students aware of appropriate websites. Some websites listed below are interactive.

Websites

<http://can-do.com/uci/ssi2001/blood-heart.html>

<http://sln.fi.edu/biosci/heart.html> (virtual labs)

http://www.gen.umn.edu/faculty_staff/jensen/1135/webanatomy/wa_cvs/wa_heart_internal1.html

<http://biology.about.com/library/organs/heart/blheart.htm>

<http://www.learn.co.uk/default.asp?WCI=Unit&WCU=657>

<http://www.gisbornesc.vic.edu.au/home/jans/home/hpluck.htm> (pluck dissection)

<http://andrewhitchcock.org/gallery/album03> (pluck images)

Experiment 5: Dissection of a rat

Suggestions for teachers

- Students should have background knowledge of the various systems of the human body so that they can identify and compare with the systems of the rat.
- Show charts and videos on dissecting rats.
- A demonstration of the dissection before students begin their own dissections would be helpful.
- Order a variety of rats; male and pregnant and non-pregnant females.
- Provide detergent, soap and a bag to dispose of the rats after dissection.
- Make the students aware of appropriate websites.

Websites

<http://www.spflrc.org/user/rats/Teeth.htm>

<http://faculty.fmcc.suny.edu/mcdarby/Pages/Lab%20Exercises/Rat.htm>

<http://www.utm.edu/~rirwin/RatAnat.htm>

<http://www.gisbornesc.vic.edu.au/home/jans/home/rat.htm>

http://www.umanitoba.ca/faculties/science/biological_sciences/lab15/biolab15_4.html

<http://www.howe.k12.ok.us/~jimaskew/ratkey.htm>

Experiment 6: Digestive enzymes

Suggestions for teachers

- Review the parts of the digestive system and digestive enzymes.
- Provide reference books, charts and models of the digestive system.
- Make the students aware of pH changes in different parts of the digestive tract.
- Make the students aware of appropriate websites.

Websites

<http://www.zoology.ubc.ca/~biomania/biol153/labfinal/digestL.doc>
<http://web.ukonline.co.uk/webwise/spinneret/nutrition/amylex.htm>
<http://web.ukonline.co.uk/webwise/spinneret/nutrition/digesy.htm>
[http://www.learn.co.uk/default.asp?WCI=Unit&WCU=2377-\(animated\)](http://www.learn.co.uk/default.asp?WCI=Unit&WCU=2377-(animated))
http://bioweb.usc.edu/courses/2003-spring/documents/bisc307-Digestive_Enzymes.pdf

Experiment 7: Asexual reproduction in plants and animals

- Students should have background knowledge on different types of reproduction in plants and animals.
- Students should know the differences between runners, rhizomes, corms, bulbs and tubers.
- Students should be aware of the advantages and disadvantages of sexual and asexual reproduction.
- If your school has a shade house, grow the plants in the shade house.
- Make the students aware of appropriate websites.

Websites

<http://biology.about.com/library/weekly/aa090700a.htm>
<http://www.learn.co.uk/default.asp?WCI=SubUnit&WCU=19638>
<http://courses.forestry.ubc.ca/frst200/lectures/VegetativeReproduction.htm#l>
<http://www.learn.co.uk/default.asp?WCI=SubUnit&WCU=19637>
http://library.thinkquest.org/22016/contribute/asex_sex.htm

Experiment 8: Food chains, food webs and food pyramids

Suggestions for teachers

- Students should have a thorough background knowledge of food chains, food webs, food pyramids and biological magnification.
- Students should construct different food chains before attempting to construct a food web. Remind them that the arrows point from the prey to the predator.
- Students could work either individually or in groups to construct the food web. Have the class compare their food webs with others.
- If students have access to the Internet, have them obtain pictures of animals and plants to include in their food webs. Make students aware of appropriate websites.

Websites

<http://www.elmhurst.edu/~chm/onlcourse/chm110/outlines/foodenergy.html> (food pyramids)
[http://www.learn.co.uk/default.asp?WCI=Unit&WCU=2400\(interactive\)](http://www.learn.co.uk/default.asp?WCI=Unit&WCU=2400(interactive))
<http://users.rcn.com/jkimball.ma.ultranet/BiologyPages/F/FoodChains.html> -(food chains and food webs)
http://www.arcytech.org/java/population/facts_foodchain.html - (food chains, food webs and food pyramids)
<http://www.marietta.edu/~biol/102/2bioma95.html> (biomagnification)

Experiment 9: Surveying an ecosystem

Suggestions for teachers

- Before the field trip, have the students practise soil testing, transect and quadrant sampling on the school grounds. Help them to use field guides to identify plant species.
- After the field trip, pool the class data and discuss the similarities and differences between the school sampling and the field trip sampling.
- Review botany books and field guides to familiarise students with Latin names and how to identify plants.
- Show students how to draw profile diagrams.
- A plant press or a heavy book may be used to preserve plant specimens.
- Make students aware of appropriate websites.

Websites

<http://fire.biol.wvu.edu/anderson/projects/01veg/>
<http://www.offwell.free-online.co.uk/4howto.htm>
http://www.tuhssd.k12.az.us/Corona_del_Sol_HS/Departments/Science/size.html
http://www.tuhssd.k12.az.us/Corona_del_Sol_HS/Departments/Science/calculations.html

Background

Timing: 2 × 45 minute lessons

Taxonomy is the branch of biology dealing with the identification and naming of organisms. **Classification** is the arrangement of objects, ideas, or information into groups, the members of which have one or more characteristics in common. Scientific classification groups all plants and animals on the basis of certain characteristics they have in common. During the 1700s, Swedish botanist Carolus Linnaeus classified all the organisms that were known at that time into two large groups: the kingdoms **Plantae** and **Animalia**. Linnaeus also developed the concept of binomial nomenclature whereby scientists speaking and writing different languages could communicate clearly. In 1969, Robert Whittaker proposed five kingdoms: **Plantae**, **Animalia**, **Fungi**, **Protista**, and **Monera**.

There are seven traditional major levels in a classification scheme. The **kingdom** is the group that contains the largest number of different organisms. Kingdom is divided into further levels of **phylum**, **class**, **order**, **family**, **genus** and **species**. Species can be defined as a group of organisms that is capable of interbreeding and producing fertile offspring under natural conditions.

The animal kingdom is divided into **vertebrates** and **invertebrates** depending on whether they have vertebral columns or not. Vertebrates are divided into five main groups: fish, amphibians, reptiles, birds and mammals. Invertebrates are divided into the following phyla: porifera, cnidaria, platyhelminthes, nematoda, annelida, mollusca, arthropoda and echinoderms.

Aim To classify animals into phyla and classes by looking at their external and internal characteristics.

Equipment

Preserved and live specimens

- | | | | | |
|---------------------------|---------------------|----------|-------------|-------------|
| • biology reference books | • stereo microscope | • snake | • jellyfish | • centipede |
| • charts | • shark | • turtle | • flatworm | • crab |
| • slides | • goldfish | • duck | • roundworm | • butterfly |
| • hand lens | • frog | • rat | • earthworm | • starfish |
| | • salamander | • bat | • snail | |
| | | • sponge | • millipede | |

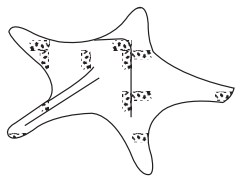
SAFETY RULES

- ⚠ Wear gloves when handling live animals. When you have finished with them, return them to the designated area.
- ⚠ Wash your hands with detergent after handling specimens.
- ⚠ Pack up the preserved specimens and photos as instructed by your teacher.
- ⚠ Use a hand lens or a stereo-microscope to observe the minute details of the organisms.

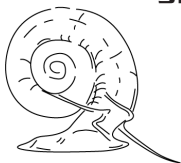
Part A – Classification of vertebrates

Method

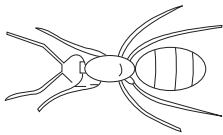
- 1 Observe each vertebrate specimen provided. Note the following characteristics.
 - skeleton: internal or external, backbone or no backbone
 - body covering: scales, hair, skin or feathers
 - symmetry (see the table below): radial, bilateral, asymmetrical or part-spiral
 - how they breathe: gills, lungs or skin
 - how they move: number of legs, fins, wings
 - body temperature: endothermic (cold-blooded) or ectothermic (warm-blooded)
- 2 Record your observations in Table 1. Classify each organism into a class: fish, amphibians, reptiles, birds or mammals.



Radial symmetry: a circular plan that can be divided into symmetrical halves along a number of vertical planes.



Part-spiral symmetry: the body is part of a spiral.



Bilateral symmetry: only one line can bedrawn to divide it into two mirror-image halves



Asymmetrical: has no symmetry, cannot be divided into equal parts.

Types of symmetry

Results

Table 1 Characteristics of vertebrates

Structures observed	Shark	Goldfish	Frog	Salamander	Snake
Backbone (present or absent)					
Skeleton (internal or external)					
Symmetry (radial, bilateral spiral or asymmetrical)					
Body covering (skin, fur, hair, feathers or scales)					
How they move (legs, wings or fins)					
How they breathe (gills, lungs or skin)					
Ectothermic or endothermic					
Class					

Structures observed	Turtle	Duck	Rat	Bat
Backbone (present or absent)				
Skeleton (internal or external)				
Symmetry (radial, bilateral spiral or asymmetrical)				
Body covering (skin, fur, hair, feathers or scales)				
How they move (legs, wings or fins)				
How they breathe (gills, lungs or skin)				
Ectothermic or endothermic				
Class				

Analysis of results

Vertebrates

- 1 What is the major difference between a vertebrate and an invertebrate?
- 2 Give examples of vertebrates that have: fins, gills, scales, feathers, legs, lungs, hair, wings.
- 3 Briefly describe the characteristics of: fish, amphibians, reptiles, birds, mammals.



Part B – Classification of invertebrates

Method

- 1
- Observe each invertebrate specimen provided. Note the following characteristics.
- skeleton: internal skeleton or external shell, backbone present or absent
- type of skin: smooth or spiny
- symmetry: radial, bilateral, asymmetrical, part spiral
- segmentation: many segments, 2 or 3 segments, not segmented
- shape of the body: flat, irregular, rounded, umbrella-shaped
- how they breathe: gills or lungs or skin
- number of jointed limbs: 4 legs, more than 4 legs, no legs
- wings: 1 or 2 pairs of wings, no wings
- body temperature: endothermic (cold-blooded) or ectothermic (warm-blooded)
- other parts of body: tentacles, antennae, arms
- 2
- Record your observations in Table 2. Classify each organism into their phyla: porifera (sponges), cnidaria, platyhelminthes (flatworms), nematoda (roundworms), annelida (segmented worms), mollusca, arthropoda or echinodermata.

Results

Table 2 Characteristics of invertebrates

Structures observed	Sponge	Jellyfish	Flatworm	Roundworm	Earthworm	Snail
Presence of back bone						
Shell or spiny skin						
Symmetry (radial, bilateral or asymmetrical)						
Number of segments						
Shape of body						
Number and type of legs						
Wings						
Tentacles or antennae						
Ectothermic or endothermic						
Phylum						

Structures observed	Millipede	Centipede	Butterfly	Crab	Starfish
Presence of back bone					
Shell or spiny skin					
Symmetry (radial, bilateral or asymmetrical)					
Number of segments					
Shape of body					
Number and type of legs					

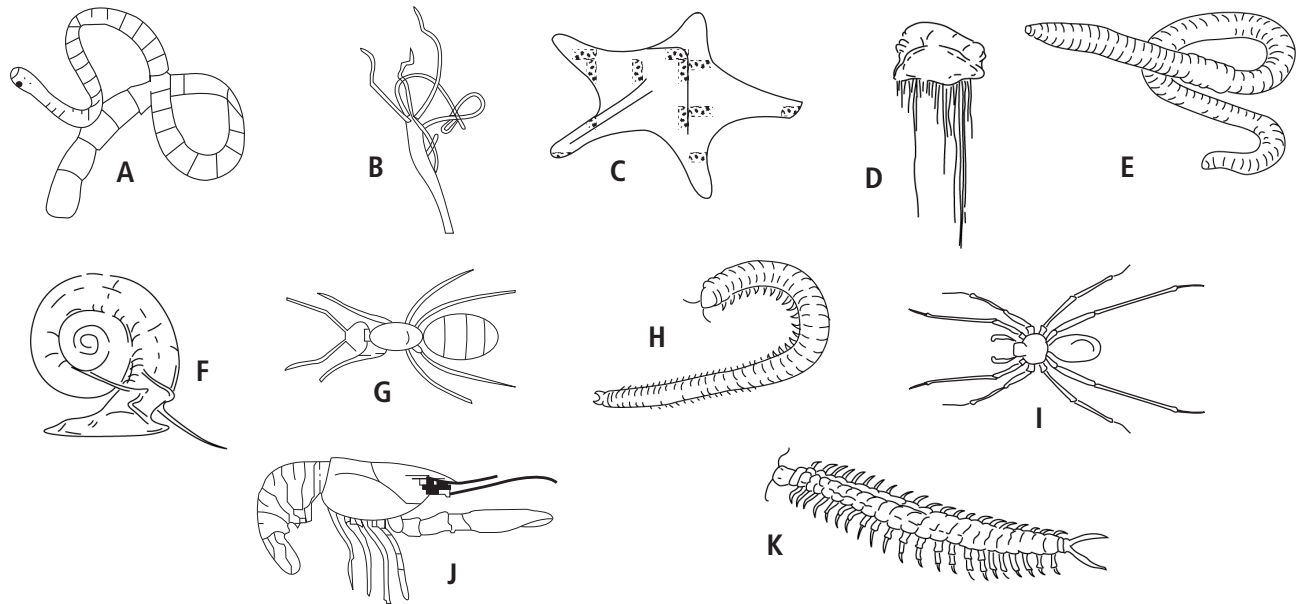
(Table continued over page)



Structures observed	Millipede	Centipede	Butterfly	Crab	Starfish
Wings					
Tentacles or antennae					
Ectothermic or endothermic					
Phylum					

Analysis of results

- 1 Why are these animals placed in the phylum invertebrata?
- 2 Give examples of invertebrates that have each of the following structures: external shell, asymmetrical, bilateral symmetry, radial symmetry, segmented, not segmented, wings, many legs.
- 3 Use the diagrams below to identify each organism and the phylum to which it belongs. List the characteristics that helped you to identify them.



Number	Name	Characteristics that helped identification	Phylum
A			
B			
C			
D			
E			
F			
G			
H			
I			
J			
K			

Conclusion

Summarise the differences between vertebrates and invertebrates.

Timing: 2 × 45 minute lessons

Background

The original purpose of biological classification was to organise the vast number of known plants into categories that could be named, remembered and discussed. **Kingdoms** are divided into a **hierarchical** system of categories called **taxa** (singular: taxon). The taxa are: **phylum** (usually called **division** in botany), **class**, **order**, **family**, **genus**, and **species**. Intermediate divisions, such as **suborder** and **superfamily**, are sometimes added to make necessary distinctions. The lower a taxon is in the hierarchy, the more closely related its members. The **species** is the fundamental unit of classification. It consists of populations of genetically similar interbreeding or potentially interbreeding individuals. The **genus** (plural: genera) is a grouping of similar, closely related species.

Green plants include all organisms commonly known as green algae and land plants. These include liverworts, mosses, ferns and other non-seed plants, and seed plants. Plants are divided into two main groups according to whether or not they have **vascular tissue**. This is the main tissue that transports water (xylem) and food (phloem) for the plant. It is usually found when plants become more specialised, such as conifers and flowering plants. The major phyla of the Plant kingdom are **bryophyta** (mosses and liverworts), **filicophyta** or **pteropsida** (ferns), **coniferophyta** (conifers), **angiosperms** (flowering plants) and **gymnosperms** (non-flowering plants). Angiosperms are again divided into **monocotyledons** (have one seed leaf) and **dicotyledons** (have two seed leaves). **Fungi** are plants which have no chlorophyll. They are placed in a separate kingdom, the Fungi kingdom.

Aim To classify plants into phyla and classes by observing their external and internal characteristics.

Equipment

- biology reference books
- specimens and photos of different types of vascular and non-vascular plants: moss, liverwort, fern, pine, geranium, lily, bean, corn, lichen
- magnifying glass
- stereomicroscope

SAFETY RULES

- ⚠ Wash your hands with soap after the activity.
- ⚠ Return the specimens to the allotted containers.
- ⚠ Use magnifying glasses or stereomicroscope to observe minute details.

Characteristics of plants in different phyla

Phylum	Common name/ examples	No. of species	Characteristics
Bryophyta	Liverworts, hornworts, mosses	24 000	Small plants living in moist habitats (the sperm must swim through water to reach the eggs). Reproduce by spores.
Psilophyta	Whiskferns	12	Simple vascular plants lacking true roots and, in some species, leaves. Reproduce by spores.
Sphenophyta or Equisetophyta	Horsetails, scouring rushes	20	Primarily found in moist, muddy habitats. The stems creep underground and produce erect annual or perennial stems with tiny leaves whorled into sheaves around the stem. They have jointed hollow stems and a rough, ribbed texture caused by the mineral silica. Reproduce by spores.
Filicophyta or Pteropsida	Ferns	12 000	Vascular plants which reproduce by spores. The stems are mostly creeping, large leaves (megaphylls) with branching veins. They are the most complex, diverse and abundant of the plant phyla that do not produce seeds.
Cycadophyta	Cycads	100	Evergreen perennial shrubs or trees. The stems are usually unbranched but thickened by some secondary growth. They have palm-like or fern-like compound leaves and contain symbiotic cyanobacteria in special roots.

Ginkgophyta	Ginkgo, maidenhair trees	1	The ginkgo is a tall tree with deciduous, fan-shaped leaves; the only living descendant of what was once a large group.
Coniferophyta or Pinatae	Conifers	550	By far the most familiar of the gymnosperms (plants having naked seeds). They are usually evergreen shrubs or trees with simple, needle-like leaves, spirally arranged. Commercially important for timber, pulp, turpentine and resin products.
Gnetophyta	Gnetophytes (cone-bearing desert plants)	70	Cone-bearing desert plants. They resemble flowering plants in many ways. It was once thought that they were a link between conifers and angiosperms.
Angiospermo-phyta or Magnoliophyta	Angiosperms, flowering plants	>230 000	The dominant land vegetation on earth. They include nearly every familiar tree, shrub and garden plant that produces flowers and seeds. They are characterised by the aggregation of sexual reproductive structures with specialised shoots (flowers) which typically comprise four kinds of modified leaves: sepals, petals, stamens (male organs) and carpels (female organs).

Method

- 1 Observe the pictures and live specimens of plants provided.
- 2 Note if the organisms have the following external and internal features.
 - Stem: Note if the stem is woody or herbaceous, hollow or solid.
 - Roots: note if the roots are fibrous or long and divided into secondary roots and root hairs.
 - Leaves: note the shape and size of the leaves.
 - Veins: note if the veins are parallel or divided into branches (net-like).
 - Flowers: note the number of petals and sepals.
 - Spores: observe the spores of ferns with a magnifying glass or a stereo microscope.
 - Cones: note the seeds in the cones.
 - Seeds: note the seed cover and the number of seed leaves (cotyledons).
- 3 Record your observations and classify each plant into its phylum or group in Table 1. Use your reference books as required.

Results

Table 1 Characteristics of plants

Structures observed	Moss	Liverwort	Lichen	Fern	Pine
Stem (herbaceous, woody hollow or solid)					
Roots (tap root or fibrous roots)					
Leaves (long, narrow, flat or needle-shaped)					
Veins (net-like or parallel)					
Vascular or non-vascular					
Flowers (5 or 6 petals)					
Cones					
Spores					
Seeds (1 or 2 cotyledon)					
Phylum or class					



Structures observed	Geranium	Lily	Bean	Corn
Stem (herbaceous, woody hollow or solid)				
Roots (tap root or fibrous roots)				
Leaves (long, narrow, flat or needle-shaped)				
Veins (net-like or parallel)				
Vascular or non-vascular				
Flowers (5 or 6 petals)				
Cones				
Spores				
Seeds (1 or 2 cotyledon)				
Phylum or class				

Analysis of results

- 1 What is the major difference between vascular and non-vascular plants?
- 2 Which plants have a vascular system?
- 3 Which plants do not have a vascular system?
- 4 Why are lichens difficult to classify?
- 5 What are the differences between the following groups of plants?
 - a Angiosperms and gymnosperms
 - b Monocots and dicots
 - c Bryophytes and filicophytes

Conclusion

Summarise how plants are classified.

CONSTRUCTION OF A DICHOTOMOUS KEY FOR ARTHROPODS



Background

Timing: 2 × 45 minute lessons

Dichotomous keys are constructed to identify known and unknown animals, plants and other organisms. This is based on a series of choices between alternative characteristics. For example, noting whether animals have vertebrae or do not in order to divide animals into two main groups. Similarly, plants can be divided into two groups by noting whether they have vascular systems or do not.

A dichotomous key for **arthropods** can be constructed by looking at characteristics such as the presence or absence of an external shell or exoskeleton, the number of legs and body segments. Arthropods are joint-footed animals that possess a hard, armour-like outer skeleton. Arthropods comprise about 95% of all known animal species, and all but 5% of arthropods are **insects**. Non-insect arthropods include the class **Crustacea** (water-dwelling crabs, shrimp, crayfish and lobsters), and land-dwelling **isopods** or sowbugs; the class **Diplopoda** (many-legged, worm-like millipedes); the class **Chilopoda** (centipedes); the class **Arachnida** (mites, ticks, spiders, scorpions, and harvestmen), the class **Xiphosura** (horseshoe crabs); and the extinct class **Trilobita** (trilobites).

Aim To construct and use a dichotomous key for known arthropods.

Equipment • Preserved or live specimens of spiders, crabs, butterflies, millipedes and centipedes

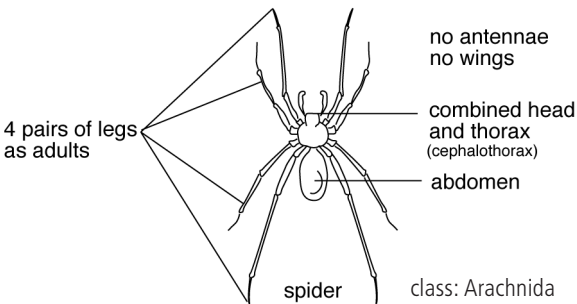
SAFETY RULES

- ⚠ Wear gloves and laboratory jackets during this activity.
- ⚠ Treat live specimens with care and return them to the specified containers.
- ⚠ Use a magnifying glass or stereomicroscope to observe minute details.
- ⚠ Pack up the preserved animals as directed by your teacher.
- ⚠ Wash your hands with soap after finishing this activity.

Basic features of arthropods

The classes and orders of arthropods can be distinguished from one another through the characteristics used in the following keys. For example, the class **Insecta** is distinguished from other arthropod classes by the following characteristics.

- 1 exoskeleton divided into three parts
- 2 three pairs of legs
- 3 one pair of antennae
- 4 compound eyes
- 5 many adults have wings



External body features that distinguish the four main classes of arthropods

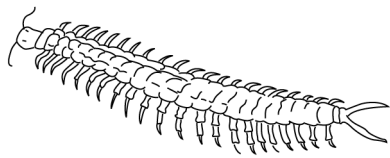
Class	Main body regions	Pairs of legs	Pairs of antennae	Wings
Crustacea	Two: cephalothorax* and abdomen (some with head and trunk)	Five or more	Two	Absent
Myriapoda	Two: head and trunk	Many: one or two pairs per trunk segment	One	Absent
Arachnida	Two: cephalothorax* and abdomen	Four	None (though palps may resemble antennae or legs)	Absent
Insecta	Three: head, thorax and abdomen	Three	One	Usually present (but many wingless forms)

*cephalothorax = fused head and thorax

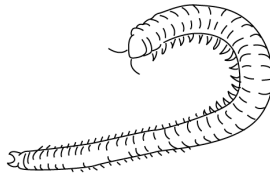
A key to major classes of adult arthropods

- 1a Typically with three pairs of legs, with or without wings, and three distinct body regions – **insects (Insecta)**.
- 1b More than three pairs of legs and wings never present – **go to 2.**
- 2a Has four pairs of legs, no antennae and the head and thorax are fused into one large cephalothorax – **arachnids (Arachnida)**.
- 2b More than four pairs of legs present – **go to 3.**
- 3a The body is broadly oval with a stout, spine-like tail – **horseshoe crabs (Xiphosura)**.
- 3b Body not as above – **go to 4.**
- 4a Usually has five to seven pairs of legs and two pairs of antennae – **crustaceans (Crustacea)**.
- 4b Has more than ten pairs of legs and only one pair of antennae; the body is long and worm-like – **go to 5.**
- 5a The body is cylindrical; there are two pairs of legs on each segment; the antennae are short – **millipedes (Diplopoda)**.
- 5b The body is flattened; there is one pair of legs per segment; the antennae are long – **centipedes (Chilopoda)**.

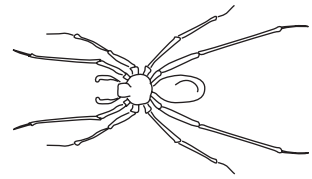
Examples of Arthropods



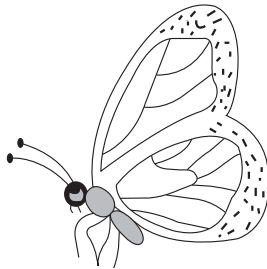
Centipede



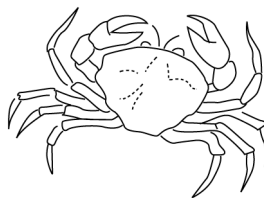
Millipede



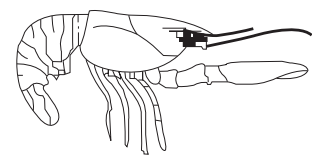
Spider



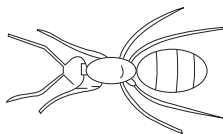
Butterfly



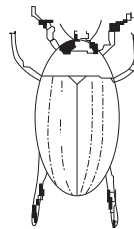
Crab



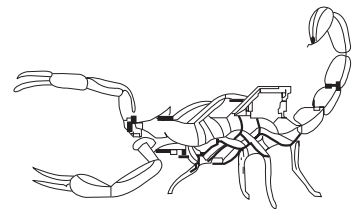
Crayfish



Ant



Beetle



Scorpion

Method

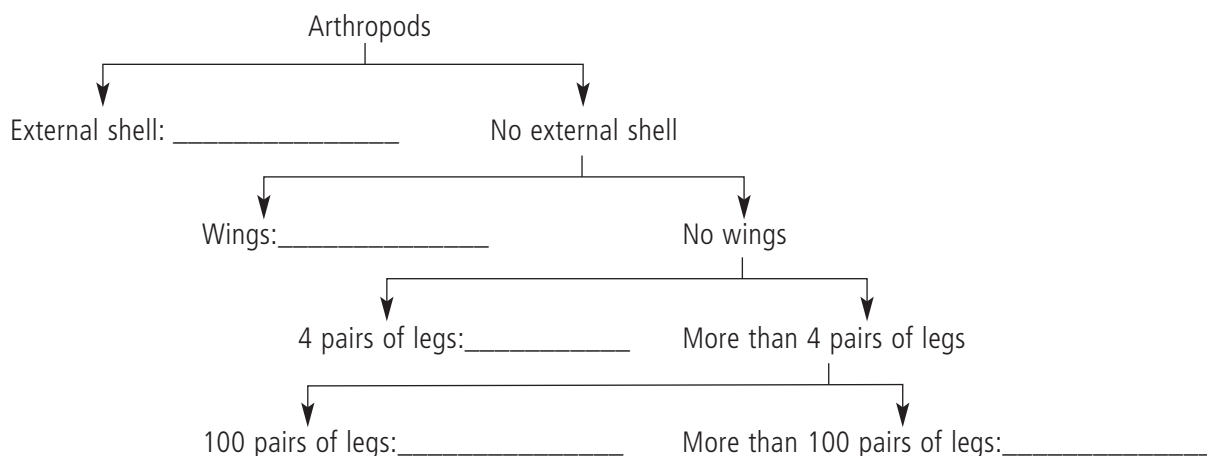
- 1 Look at the flow chart under Results. The dichotomous key as a diagrammatic key, or flow chart, is begun by dividing the arthropods into two groups according to the presence or absence of an external shell. Fill in the blank space.
- 2 The 'No external shell' group is subdivided according to the presence or absence of wings and then subdivided twice more according to the number of legs. Fill in the blank spaces.
- 3 Use the completed diagrammatic key to help you complete the dichotomous key using a normal text key.

- 4 Use different characteristics to construct two dichotomous keys of your own: one diagrammatic key (flow chart) and one normal text key. You could begin with the presence or absence of wings or with the number of legs.

Results

Dichotomous keys

• Diagrammatic key



• Text key

Arthropods

- 1a External shell – _____
- 1b No external shell – **go to 2.**
- 2a Wings. – _____
- 2b No wings – **Go to 3.**
- 3a 4 pairs of legs – _____
- 3b More than 4 pairs of legs – **go to 4.**
- 4a 100 pairs of legs – _____
- 4b More than 100 pairs of legs – _____

Analysis of results

- 1 List the distinguishing characteristics of crabs (Crustacea), butterflies (Insects), spiders (Arachnida), millipedes (Diplopoda) and centipedes (Chilopoda).

Conclusion

- 1 Why is a dichotomous key important for identifying organisms?
- 2 Can you use the same dichotomous key if you are asked to identify an ant, a crayfish, a scorpion and a beetle? Explain.
- 3 If you were to find an arthropod that has no wings, three pairs of legs and a body divided into three segments, in which group would you place it? Explain.

DISSECTION OF A SHEEP'S PLUCK



Background

Timing: 2 × 45 minute lessons

The sheep's pluck consists of the lungs, heart and liver. The **heart** is divided into four chambers: **right atrium**, **right ventricle**, **left atrium** and **left ventricle**. The right side of the heart is separated from the left side by a thick muscular wall called a **septum**. The left atrium-ventricle valve is known as the **bicuspid** or **mitral valve**. The right atrium-ventricle valve is known as the **tricuspid valve**. This valve allows blood to flow only in one direction: from the atria to the ventricles. The right atrium receives deoxygenated blood from the body via the **superior vena cava** and the **inferior vena cava**. The blood then flows to the right ventricle and from there it is carried to the lungs by the **pulmonary arteries**.

The **lungs** are pale pink in colour and feel firm and spongy. The left lung is divided into two lobes: the upper lobe and the lower lobe. The right lung is divided into three lobes: the upper lobe, middle lobe and lower lobe. When humans breathe in, air enters through the nostrils and goes to the **pharynx**, **larynx**, **trachea**, **bronchi** and **bronchioles** and finally to small sacs called **alveoli**. Exchange of gases takes place by diffusion between the blood in the capillaries and air in the alveoli. Oxygen rich blood is carried back to the left atrium by the **pulmonary veins**. The blood then enters the left ventricle and is pumped out of the heart via the **aorta**.

The **liver** is a reddish brown organ which is divided into two lobes: the right and the left. The right lobe is about six times the size of the left. The **gall bladder** is located in the liver and stores bile, which helps in emulsifying fats. The **hepatic portal vein** carries blood from the liver to the heart.

Aim

To dissect and identify the organs present in a sheep's pluck, and to compare them with human organs.

Equipment

- sheep's pluck containing heart, lungs and liver with blood vessels intact
- scalpel
- sharp scissors
- forceps
- probe
- dissecting board
- surgical gloves
- newspaper
- 30-cm clean rubber tubing to fit the trachea
- piece of string
- models of the human torso and heart
- 250 mL beaker of cold water

SAFETY RULES

- ⚠ Wear a lab coat and surgical gloves when dissecting.
- ⚠ Be careful not to cut your fingers while using the dissecting tools.
- ⚠ The tools of dissection are dangerous. Make sure you are prepared and not distracted when you perform dissections.
- ⚠ Organs should be placed in a bag for disposal. No organs or tissues should be left in the sinks.
- ⚠ Clean the sinks with detergent after the experiment is completed.
- ⚠ Dispose of your gloves in the rubbish bin and wash your hands with disinfectant.

Method

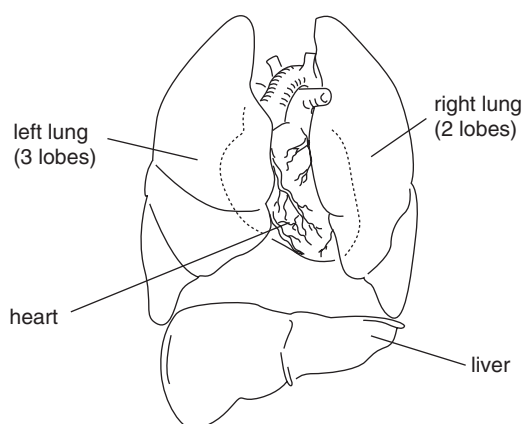


Figure 1: The pluck of the sheep

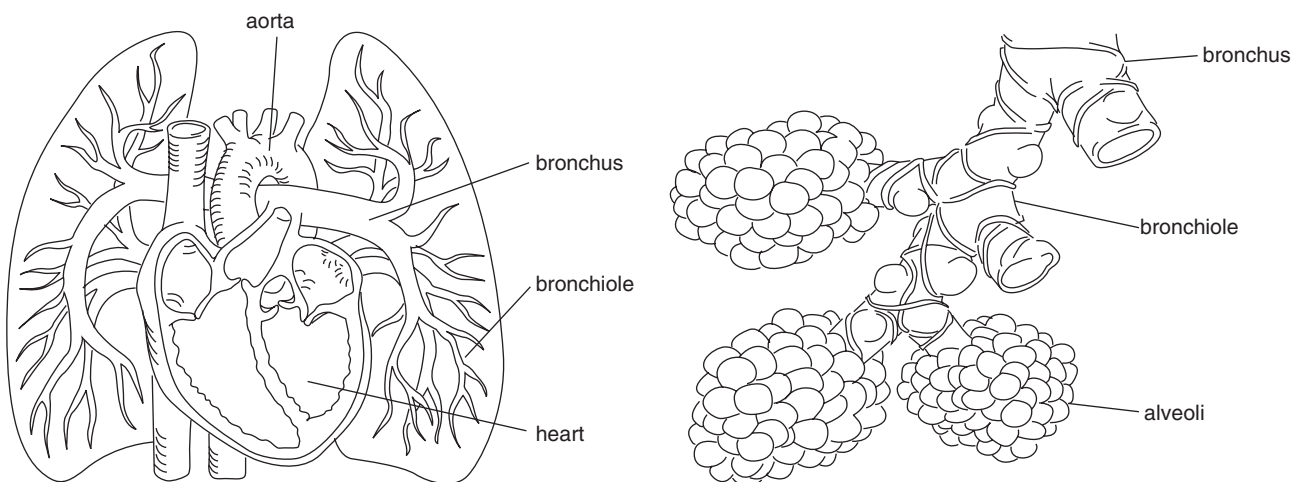


Figure 2: Parts of the lungs

Lungs

- 1 Wear surgical gloves. Arrange the pluck on a dissecting board so that the heart is on top of the lungs. Identify the structures labelled in Figure 1.
- 2 Insert the rubber tubing through the **trachea** (wind pipe) and tie it tightly with a piece of string. Blow air into the tube and observe if the lungs inflate. **Note:** If the lungs are cut, they will not inflate.
- 3 Note the colour and the number of lobes in each lung.
- 4 Remove the rubber tube and feel the cartilage rings around the trachea.
- 5 Use sharp scissors to cut the trachea on one side. Trace the **bronchus** and **bronchioles** in both lungs. Observe the **alveoli** at the end of the smallest bronchiole.
- 6 Observe the **pulmonary arteries** coming from the right ventricle of the heart to the lungs, and the **pulmonary veins** going from the lungs to the left atrium of the heart.
- 7 Cut a small piece of the lung tissue and use forceps to place it in water. Observe whether the lung tissue floats or sinks.
- 8 Remove the lungs and place them on newspaper.

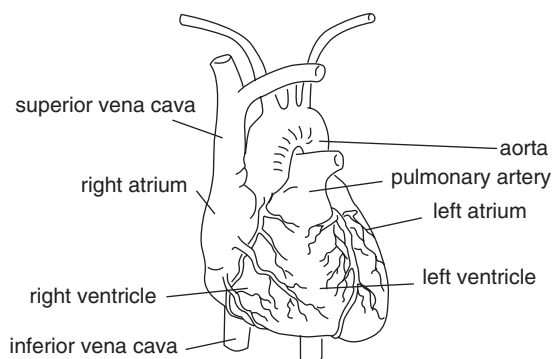


Figure 3: External structure of the heart

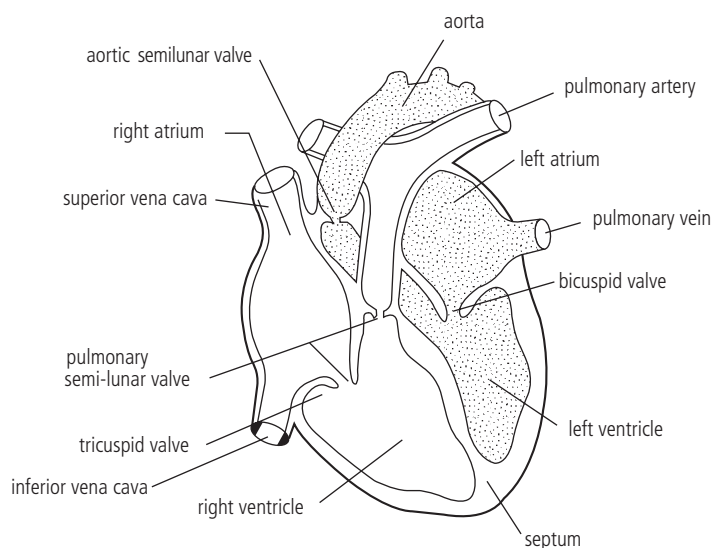


Figure 4: Internal structure of the heart

Heart

- 9 Position the heart as shown in the figure 3. Observe its shape and colour and then identify each part. Remember that the veins have thinner walls than the arteries. **Note:** Parts of the atria and blood vessels may have been cut by the butcher.
- 10 Locate the right atrium with your thumb by squeezing the heart and feeling for the thin-walled chamber.
- 11 Make a shallow cut (not more than 5 mm deep) from the right atrium, down the right ventricle, to the apex of the heart.
- 12 Continue the shallow cut around the apex and up the left ventricle so that there is a line dividing the ventricles into two equal halves. Slice along the shallow cut to divide the ventricles in half. Check to see if the walls are the same thickness in each **ventricle**.
- 13 Make an incision in the wall of the left atrium. Gently push the handle of the probe into the incision and feel the **bicuspid** valve, which divides the left atrium from the left ventricle. The bicuspid valve has two thin, sail-like, transparent structures that are held by strong, white, chord-like **chordae tendinae**.
- 14 Carefully push the handle of your probe through the valve into the **left ventricle**. Explore the structure and operation of this valve.
- 15 Use the handle of your probe inside the **left atrium** to find all four **pulmonary veins**.
- 16 Find where the **aorta** leaves the left ventricle and carefully push the handle of your probe into it. Note the location where the probe exits the heart.
- 17 Remove the probe, and then gently push the handle in the opposite direction from the exit of the aorta towards the left ventricle. At a point close to the left ventricle, the probe will be stopped by a valve called the **aortic semi-lunar valve**.
- 18 Leave the probe in place and make an incision to find the aortic semi-lunar valve.
- 19 Make an incision in the wall of the right atrium. Gently push the handle of the probe into this incision and locate the **tricuspid valve**, which separates the right atrium from the right ventricle. The tricuspid valve is similar to the bicuspid valve except that it has three sail-like flaps instead of two.
- 20 Use the handle of your probe inside the right atrium to find the **superior vena cava** and the **inferior vena cava**.
- 21 Carefully push the handle of your probe into the pulmonary artery and notice the location where the probe exits the heart.
- 22 Remove the probe, then gently push the handle in the reverse direction from the exit of the pulmonary artery towards the right ventricle. At a point close to the right ventricle, the probe will be stopped by a valve called the **pulmonary semi-lunar valve**. Leave the probe in place and make an incision to find the valve.
- 23 Feel the thickness of the right atrium, right ventricle, left atrium and left ventricle.
- 24 Observe the **coronary arteries** supplying blood to the heart muscles.

Liver

- 25 Observe the shape, colour and texture of the liver.
- 26 Observe the greenish-coloured gall bladder and the bile duct.
- 27 Remove the liver and place it on the newspaper with the lungs.
- 28 Dispose of all the organs by wrapping them in newspaper and then in a rubbish bag.
- 29 Wash the dissecting instruments and board with detergent and dry them.
- 30 Remove your gloves and wash your hands with soap.

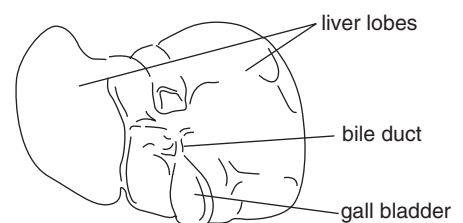


Figure 5: Liver

Results

- 1 Draw and label the structures of the lungs and liver.
- 2 Draw and label the external and internal structures of the heart.

Analysis of results**Lungs**

- 1 Describe the colour and structure of the lungs.
- 2 What is the function of the cartilage rings around the trachea?
- 3 What happened when the lung tissue was placed in water? Explain.
- 4 What are the functions of the alveoli?

Heart

- 5 Name the chambers of the heart.
- 6 Which side of the heart is thinner?
- 7 Which chamber of the heart is thickest?
- 8 Why do the ventricles have thicker walls than the atria?
- 9 What are the functions of the bicuspid and tricuspid valves between the atria and ventricles?
- 10 Which blood vessel is the largest in the body?
- 11 Compare the dissected sheep's heart with a model of human heart. Explain the similarities and differences.
- 12 What are the functions of semi-lunar valves situated between the pulmonary artery and the right ventricle?
- 13 Describe the functions of the each of the following. (Refer to your text book as required.)

Structures/blood vessels	Functions
Right atrium	
Right ventricle	
Left atrium	
Left ventricle	
Gall bladder	
Inferior venacava	
Superior venacava	
Pulmonary arteries	
Pulmonary veins	
Coronary arteries	
Aorta	

Liver

- 14 Describe the structure of the liver.
- 15 Describe the appearance of the gall bladder.

Conclusion

Look at a model of the human torso and compare the human organs (heart, lungs and liver) with the sheep organs.

DISSECTION OF A RAT



Background

Timing: 2 × 45 minute lessons

Dissection is a technique used to study the **internal organs** of a dead animal. It shows the internal structure of the animal, how the tissues look and feel, and how they are interrelated. All mammals possess similar organs and systems whether they are small or large in size. To understand the structure and functions of our own systems, it is helpful to dissect smaller mammals such as rats.

As rats are herbivorous, they have four long, sharp incisors for gnawing and twelve molars for chewing. They have no canines or premolars. The toothless space between the incisors and molars is called the **diastema**. Their digestive system is similar to that of humans except that they have a proportionally longer small intestine, a large **caecum** to digest cellulose and no gall bladder. There are no differences between human and rat excretory systems. The female rats have a V-shaped uterus that can hold many foetuses, whereas the human uterus is smaller and usually holds only one foetus at a time.

Aim

To dissect and note the structures and functions of various organs and systems in the rat.

- Equipment**
- a preserved rat
 - dissecting needle
 - small ruler
 - scalpel
 - dissecting board
 - model of a human torso
 - sharp scissors
 - surgical gloves
 - charts and videos on rat dissection
 - dissecting probe
 - long string
 - newspaper
 - forceps
 - drawing pins

SAFETY RULES

- ⚠ Wear a lab coat and surgical gloves when dissecting.
- ⚠ The rat is stored in a chemical that can be painful and harmful if you get it in your eyes, nose or mouth.
- ⚠ The tools of dissection are dangerous. Make sure you are prepared and not distracted when you perform dissections.
- ⚠ Dispose of the dissected rat by wrapping it in newspaper and then put it in the bag provided. Do not discard any organs in the sink.
- ⚠ Clean the sinks with detergent after the experiment is completed.
- ⚠ Dispose of your gloves in the rubbish bin and wash your hands with disinfectant.

Method

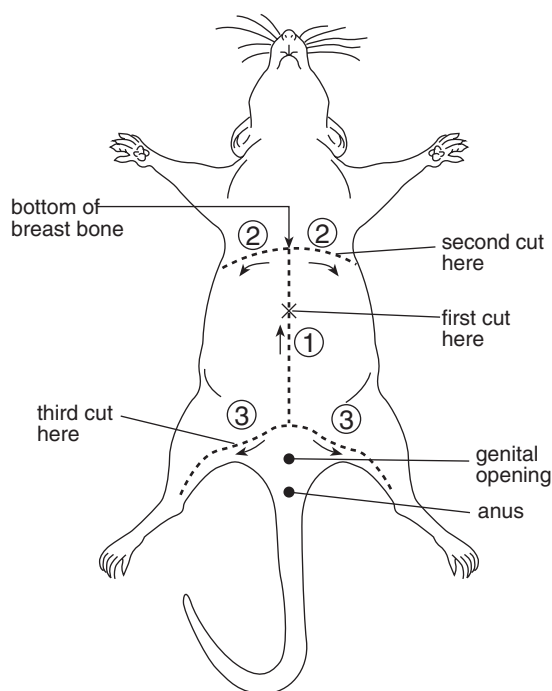


Figure 1: External diagram

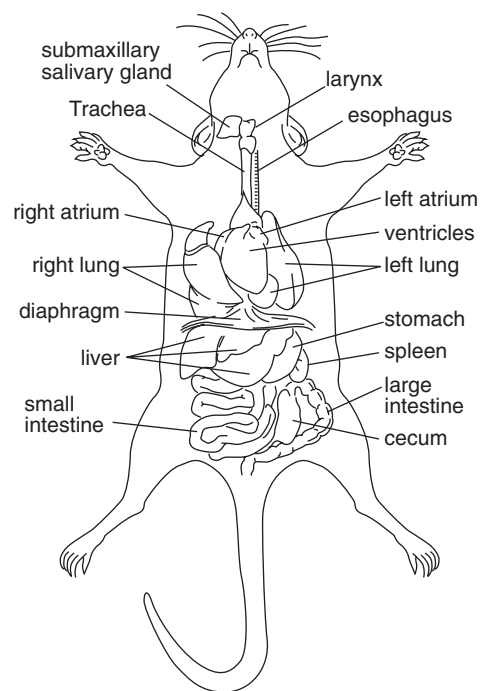


Figure 2: Internal diagram

- 1 Get the necessary dissecting equipment, dissecting board, string, drawing pins, a ruler and newspaper. Obtain a preserved rat specimen for your group.
- 2 Wearing surgical gloves, lay the rat down on its back on the dissecting board. Pin the paws to the dissecting board.
- 3 Secure the rat firmly to the dissecting board with a string. (Your teacher will show you how to do this.)
- 4 Use forceps to grab a pinch of skin between the hind legs and make a V-shaped cut with a pair of sharp scissors. Continue to cut upwards along the mid-line up to the throat. (See figure 1.)
- 5 Cut across the two front legs and the two back legs, taking care not to cut the underlying muscles. To avoid damage to the internal organs, be sure to point the blunt end of the scissors towards the rat and lift the muscles as you cut.
- 6 Peel the skin off the rat and pin the skin to the dissecting board.
- 7 Pin the muscles to the board securely. Now you will be able to see the chest cavity containing the heart, lungs and oesophagus and the abdominal cavity containing the digestive system, excretory system and reproductive system. The chest cavity is separated from the abdominal cavity by a thin muscular membrane called the diaphragm. (See figure 2.)

Chest cavity

- 8 Open the chest cavity by cutting through the centre of the ribs. Carefully extend the cut into the hollow of the throat between the bottom jaws.
- 9 Observe the lungs and the heart located in the chest cavity.
- 10 Trace the trachea as it divides and goes into the lungs.
- 11 The heart is covered by a membrane known as the pericardium. The apex of the heart is formed by the large, muscular, left ventricle. The right ventricle is not clearly distinguished externally from the left, but is smaller and has thinner walls.
- 12 Remove the lungs and the heart and place them aside on newspaper.

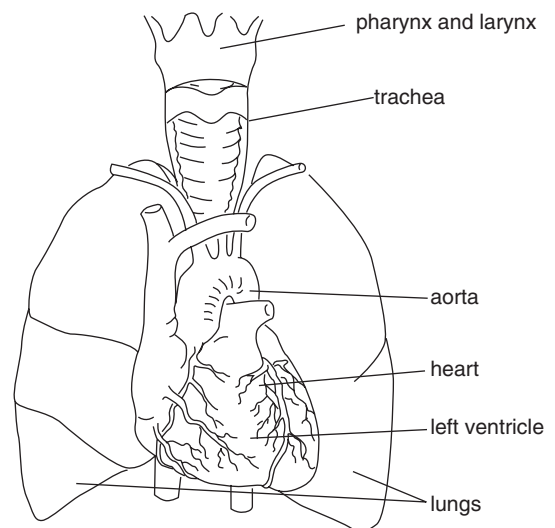


Figure 3: Chest cavity

Digestive system

• Mouth and teeth

- 13 Open the mouth of the rat. Observe the shape and size of the teeth and the empty space between the incisors and molars. Check the number of incisors and molars.

• Abdominal cavity

- 14 In the abdominal cavity, pick up the esophagus as it comes through the diaphragm and note where the esophagus enters the stomach.
- 15 Over the stomach you will see a reddish-brown liver which is divided into five lobes. The rat has no gall bladder.
- 16 Note the spleen. It is a flattened, reddish organ lying to the left and posterior to the stomach.
- 17 A glandular organ called the pancreas is found near the stomach.

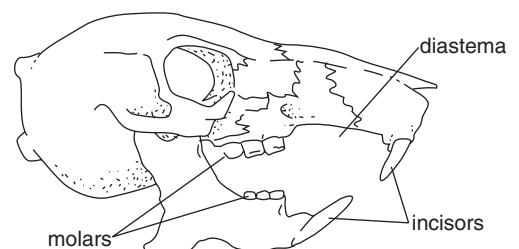


Figure 4: Rat's skull and teeth

- 18 Follow the small intestine all the way along until it becomes the large intestine.
- 19 At the junction between the small and large intestines, there is a blind pouch called the caecum which is important in cellulose digestion. Humans do not have a caecum.
- 20 At the end of the large intestine there is an expanded portion called the rectum, which opens to the exterior through the anus.
- 21 Carefully stretch the intestines and examine the thin mesentery membrane attached to the small intestine which holds the intestine together.
- 22 Use your fingers to separate the intestines from the thin membrane. Pin it to one side on the dissecting board.
- 23 Measure the diameter and length of the whole digestive system and the esophagus, stomach, liver, small intestine, large intestine and caecum.
- 24 Enter your measurements in the Table 1. Then draw and label the diagram of the digestive system as described under Results.

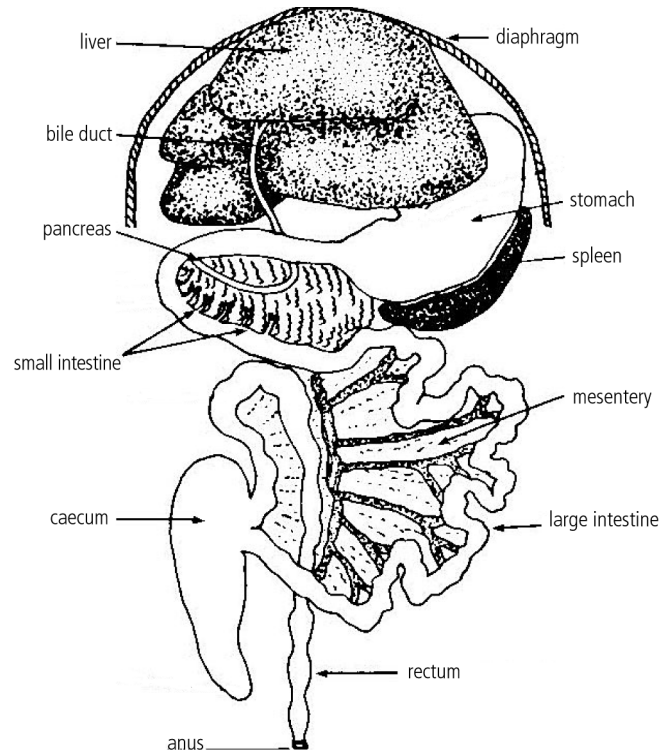


Figure 5: Rat's digestive system

Urino-genital system

• Excretory system

- 25 Wash the abdominal cavity with water and note the colour and size of the kidneys, which are located on either side of the backbone.
- 26 Trace the ureters leading from the kidney to the urinary bladder. The bladder expels urine via the urethra.

• Male reproductive system

- 27 Examine the genital area of your rat. If your rat is a male rat, observe the organs shown in figure 6.
- 28 Locate the male's scrotum and cut longitudinally through the skin to locate the testes.
- 29 Separate the skin from the testes and continue the cut up to the abdominal cavity. Notice the epididymis lying around each testis.
- 30 Find the vas deferens that leads from each testis to the urethra. Note: You will probably have to cut through some muscles in the pelvis area to locate some of these structures. Be careful as you do so.
- 31 To either side of the bladder you will find two sets of glands. The smaller, round, more posterior ones are the prostate glands. The larger pair of glands are the seminal vesicles, which are actually two different glands – the vesicular glands and the coagulating glands.

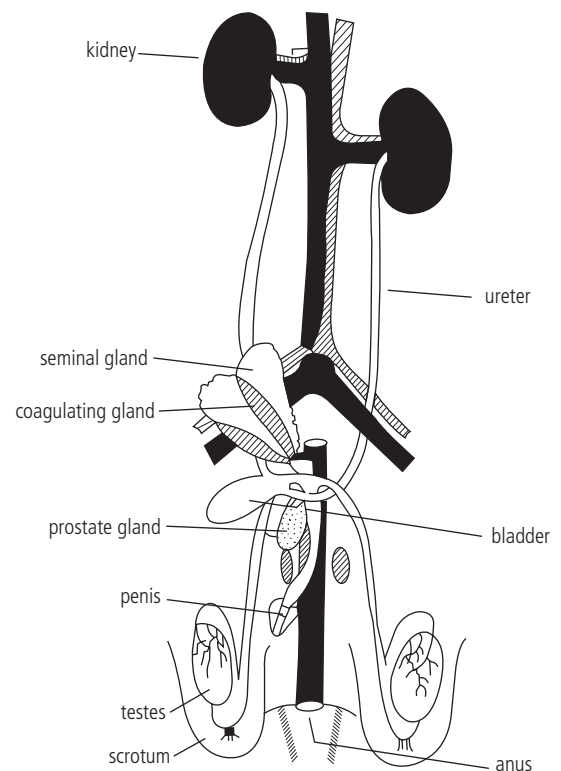


Figure 6: Male urino-genital system

- 32 Follow the urethra as it goes through the penis. In males, the bladder and the testes both empty to the outside through the urethra.

• **Female reproductive system**

- 33 If your rat is a female rat, observe the organs shown in figure 7. In the female you will find two **V-shaped uteri**. The two uteri join and open into the vagina.
- 34 At the anterior end of each uterus there is a short, convoluted **oviduct**, which opens into a transparent pocket around the small, round ovary.
- 35 Some female rats may be pregnant and have small embryos in the uteri. If your rat is pregnant, cut longitudinally along the uteri and observe the embryos.
- 36 On the exterior, just ventral to the tail, you will find the anus, the **vaginal opening** and the **urethral opening**.
- 37 Observe the reproductive organs of a female rat dissected by other groups and note the differences.
- 38 After finishing the dissection, wrap the rat and the organs in a newspaper and place it in a bag provided for disposal. Make sure no organs are left in the sink.
- 39 Clean your dissecting board and instruments with detergent and pack up.
- 40 Dispose of your gloves in the bag provided, and wash your hands thoroughly with soap.

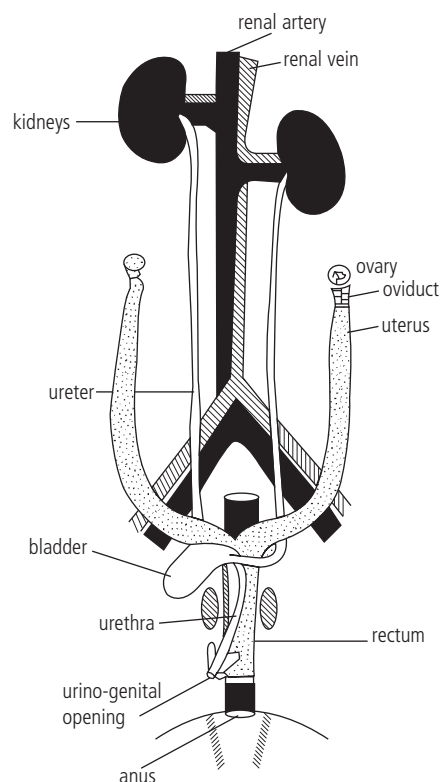


Figure 7: Female urino-genital system

Results:

- 1 Draw and label diagrams of the digestive system and the male and female urinogenital systems of the rat.
- 2 Enter your measurements of the digestive organs in Table 1.

Table 1 Measurements of digestive organs

Organ measured	Diameter (cm)	Length (cm)
Esophagus		
Stomach		
Liver		
Small intestine		
Large intestine		
Caecum		
Whole digestive system		

Analysis of results

Chest cavity

- 1 Which organs did you see in the chest cavity?
- 2 What separated the chest cavity from the abdominal cavity?

Digestive system

- 3 Describe the type of teeth possessed by the rat.
- 4 Describe the shape and structure of the stomach.
- 5 Which organ is the largest? How much of the abdominal cavity did it occupy?
- 6 What differences did you notice between the small intestine and the large intestine?
- 7 Where is the caecum situated? What is its purpose?

Urino-genital system

Excretory system

- 8 Describe the shape and position of the kidneys.
- 9 What are the functions of the ureters, urinary bladder and urethra?

Reproductive system

- 10 Describe the functions of these reproductive organs: penis, testes, ovaries, uterus.
- 11 What is the difference between a human uterus and a rat uterus?
- 12 How many foetuses did you notice in the rat?

Conclusion

Examine a model of a human torso. Describe the differences between the rat and human digestive system, excretory system and reproductive system.

Background

Timing: 2 × 45 minute lessons

Food is digested in the **digestive system** with the help of digestive **enzymes**. These enzymes are secreted by the cells lining the inner surfaces of various glands of the digestive system. **Amylase** from the salivary glands and the pancreas digest starch to maltose in the mouth and small intestine. The optimum pH for pancreatic amylase is 6.7 to 7.0. **Proteases** are enzymes that break down proteins to polypeptides. **Pepsin** is secreted in the stomach and its optimum pH is 2. **Trypsin** is secreted by the pancreas and its optimum pH is 7–8. **Fat** is digested by **pancreatic lipase** in the intestine into fatty acids and glycerol. **Bile** from the liver emulsifies fats, so that lipase has an increased surface to work on. This action takes place in the small intestine. The optimum temperature for all these enzymes is 37°C.

SAFETY RULES

- ⚠ Wear safety goggles because acids and bases can damage your eyes. If you get chemicals on your skin, wash immediately with plenty of water.
- ⚠ Use the test tube holders to pick up hot test tubes, and tongs to pick up beakers.
- ⚠ Dispose of egg whites and liquids in specified containers.
- ⚠ Do not contaminate pipettes and spatulas.

Part A – The digestion of milk fats by pancreatic lipase

Aim To investigate how pancreatic lipase and bile salts digest fats to fatty acids and glycerol.

- Equipment**
- safety goggles
 - bile salts (powder)
 - 12 mL of 0.5% pancreatic lipase
 - 10 mL cream
 - 10 mL graduated cylinder
 - 10 mL of 1% litmus solution
 - distilled water
 - pipettes
 - 6 test tubes
 - test tube rack
 - thermometer
 - 6 × 500 mL beakers
 - matches
 - thermometer
 - Bunsen burner
 - incubator set at 37°
 - hotplate
 - test tube holder
 - tongs
 - marking pens
 - spatula
 - tripod
 - gauze mat
 - heat mat

Method

- 1 Label four test tubes 1, 2, 3 and 4. Add solutions to the test tubes as shown in the table below. Add the enzyme last.

Test tube 1	Test tube 2	Test tube 3	Test tube 4
3 mL cream + a pinch of bile salts + 3 mL lipase + 3 mL litmus	3 mL cream + a pinch of bile salts + 3 mL distilled water + 3mL litmus (no lipase)	3 mL cream + 3 mL lipase + 3 mL litmus (no bile salts)	A pinch of bile salts + 3 mL lipase + 3 mL distilled water + 3 mL litmus (no cream)

Note: Litmus is a pH indicator which changes to red in acids medium and to blue in alkaline medium. The enzymatic digestion of fat (triglyceride) releases fatty acids (e.g. butyric acid) which react with litmus and change its colour to red.

- 2 Shake the test tubes to mix the contents thoroughly. Note the colour of the solution in each test tube and place the test tubes in a water bath at 37°C. Incubate for 1 to 1.5 hours at 37°.
- 3 Shake the test tubes and record their colour in Table 1 every 15 minutes until no colour change is observed. Choose terms to describe the colours such as: milky blue, milky violet, milky purple, pinkish purple, reddish purple, transparent blue, transparent red.
- 4 Record the colour change in litmus and also indicate whether the fat was digested or not.

Results

Table 1

Test tube	Presence of fatty acids (yes or no)	Rate of digestion (fast, slow or no digestion)	Colour of the litmus after 1.5 hrs	Explanation of each result
1				
2				
3				
4				

Analysis of results

- 1 What happened in each of the four test tubes? Explain.
- 2 How did you infer that digestion of fat has occurred?

Part B – Digestion of proteins by pepsin and trypsin

Aim To identify how pepsin and trypsin digest proteins to polypeptides.

Equipment

- safety goggles
- marking pen
- 8 test tubes
- test tube rack
- test tube holder
- 18 cubes of boiled egg white (protein)
- matches
- knife
- metric ruler
- 10 mL graduated cylinder
- 10 mL distilled water
- 5 mL dilute hydrochloric acid (HCl)
- 15 mL dilute sodium hydroxide (NaOH)
- tripod
- gauze mat
- heat mat
- 5 mL copper sulfate solution
- 20 mL pepsin solution
- 20 mL of trypsin solution
- incubator at 37°C
- 6 crystals of thymol
- 500 mL beaker
- tongs
- Bunsen burner

Advance preparation

Egg and solutions of pepsin and trypsin

- Boil the egg for at least 5 minutes. Rinse the egg in cool distilled water and cut the egg white into 5 mm cubes. You should get 50 cubes from 1 egg. When the egg white is boiled it becomes solid because the protein in the egg white (albumin) coagulates when heated.
- To prepare a 2% pepsin solution, place 4 g of dry pepsin in a flask and dilute it with 200 mL of distilled water. Use dilute HCl to adjust the pH of the solution to 2.
- To prepare a 5% trypsin solution, place 10 g of dry pancreatin (a mixture of trypsin, amylase, and lipase) in a flask and dilute it with 200 mL distilled water. Use dilute NaOH to adjust the pH of the solution to 8.

Method

Testing for a protein

- 1 Take 2 test tubes and label them 1 and 2.
- 2 Add 1.5 mL of sodium hydroxide and 1.5 mL of egg white in test tube 1.
- 3 Add 10 drops of copper sulfate solution and allow 5 minutes for the colour to develop. A **dark pink** or **violet colour** indicates the presence of protein.
- 4 Repeat this procedure with 1.5 mL distilled water instead of egg white in test tube 2 and note if the colour changes.

Testing for protein digestion by pepsin and trypsin

- 5 Use a wax pencil to label 6 test tubes as: C-2, C-8, P-2, P-8, T-2, and T-8.
- 6 Use the test tubes marked C for your controls. P indicates pepsin, and T indicates trypsin. The number will indicate the pH of the test tube's contents.
- 7 Place three cubes of boiled egg white in each test tube.
- 8 Add 5 mL of **distilled water** to each test tube labelled C. Add 5 mL of **pepsin** to each test tube labelled P. Add 5 mL of **trypsin** to each test tube labelled T.
- 9 Add 5 mL of **dilute hydrochloric acid** (HCl) to each tube labelled 2. Add 5 mL of dilute **sodium hydroxide** (NaOH) to each tube labelled 8. Observe the shape and colour of the egg white cubes at the start of the experiment and record your observation in Table 2.

C2	C8	P2	P8	T2	T8
3 cubes of egg + 5 mL distilled water + 5 mL HCl	3 cubes of egg + 5 mL distilled water + 5 mL NaOH	3 cubes of egg + 5 mL pepsin + 5 mL HCl	3 cubes of egg + 5 mL pepsin + 5 mL NaOH	3 cubes of egg + 5 mL trypsin + 5 mL HCl	3 cubes of egg + 5 mL trypsin + 5 mL NaOH

- 10 Place the test tubes in an incubator at 37°C and leave them overnight. Make sure the temperature remains at 37°C. Note: You can add a small crystal of **thymol** to each test tube to prevent bacterial growth.
- 11 The next day, observe the tubes and note any changes in the size and appearance of the egg white cubes in each test tube. Record your observations in Table 2. If no changes are seen after 24 hours, leave the apparatus set up for another 24 hours.

Results

Table 2

Test tube	Appearance of egg white				Explanation of results
	Before		After		
	Food size	Food appearance	Food size	Food appearance	
C-2					
C-8					
P-2					
P-8					
T-2					
T-8					

Analysis of results

- 1 What colour changes did you observe in test tubes 1 and 2 when copper sulfate was added to test for the presence of protein?
- 2 How did the test tubes with a protease (pepsin and trypsin) differ from the test tubes with no protease (control)?
- 3 At which pH did pepsin break down protein most completely?
- 4 At which pH did trypsin break down protein most completely?
- 5 According to your results, which specific protease would break down protein in the small intestine? In the stomach?
- 6 What is the significance of placing the test tubes in the incubator at 37°C?

Part C – Digestion of carbohydrates by pancreatic amylase

Aim To identify how pancreatic amylase digests carbohydrates to simple sugars, maltose and glucose.

Equipment

- 30 mL of 1% starch solution
- distilled water
- 2N hydrochloric acid
- 30 ml of 3% pancreatic amylase
- iodine solution (I_2KI)
- 30 mL of 1% glucose solution
- 30 mL of 1% maltose solution
- Benedict's solution
- 14 test tubes
- 2 test tube racks
- pH paper
- 4 × 250 mL beakers
- Bunsen burner
- tripod
- gauze mat
- heat mat
- thermometer

Method

- Take 6 test tubes and label them C1, C2, C3, C4, C5 and C6.
- Add 5 drops of **I_2KI** to test tubes C1, C2 and C3 and shake the test tube. If the iodine retains its yellow-brown colour, starch is absent. If a purple or blue-black colour appears, starch is present. The deeper the colour, the greater the amount of starch. Note the colour changes in Table 3.
- Add a small amount of **Benedict's solution** to test tubes C4, C5 and C6. Heat in a boiling water bath for 5 minutes and then cool. If the Benedict's solution retains its clear **blue colour** and no precipitate is formed, sugar is absent. The presence of sugar is indicated by the development of a precipitate, which may range in colour from green through to yellow, orange and red.

Controls

Test tube C1	Test tube C2	Test tube C3	Test tube C4	Test tube C5	Test tube C6
Starch + I_2KI	Glucose + I_2KI	Maltose + I_2KI	Starch + Benedict's	Glucose + Benedict's	Maltose + Benedict's
Blue black	Yellow	Yellow	Blue	Green	Green
Starch test			Sugar test		

- Keep these test tubes as controls to compare to the test tubes prepared for starch digestion.
- Label four test tubes A, B, C and D. Add solutions to four test tubes as given in the table below. Add the pancreatic amylase enzyme last.

Test tubes				Test tubes			
A	B	C	D	A2	B2	C2	D2
5 mL starch + 5 mL amylase	5 mL starch + 5 mL water	5 mL water + 5 mL amylase	5 mL water + 5 mL amylase + 5 drops HCl	5 mL starch + 5 mL amylase	5 mL starch + 5 mL water	5 mL water + 5 mL amylase	5 mL water + 5 mL amylase + 5 drops HCl
Starch test before incubating at 37°C				Sugar test after incubating 37°C			

- Shake the tubes and determine and record the pH of each reaction by using **pH paper**. Record the results in Table 3.
- Place 8 test tubes in a water bath at 37°C. Incubate for at least 45 minutes. Shake the tubes every 15 minutes.
- Divide each of the samples by pouring half into one of four clean test tubes, labelled A2, B2, C2 and D2.
- Test A, B, C, D test tubes for starch by adding 5 drops of **iodine solution** (I_2KI). Shake and record the colour of the iodine solution in Table 3.
- Test A2, B2, C2 and D2 test tubes for reducing sugar (maltose) by adding 5 mL of **Benedict's solution** to each, and then placing in a boiling water bath for 5 minutes. Remove and rate the precipitate for colour.
- Record the colour of the precipitate in Table 3 and compare the colour with the control test tubes.

Results

Table 3 Results of iodine and Benedict’s tests for starch digestion by amylase

Solutions used	Test tubes	pH	Starch test (colour)	Presence of starch (yes or no)	Test tubes	pH	Sugar test (colour)	Presence of maltose (yes or no)	Brief explanation of result
Starch + enzyme	A				A2				
Starch + distilled water	B				B2				
Distilled water + enzyme	C				C2				
Starch + acid + distilled water	D				D2				

Analysis of results

- 1
- Which of the 4 test tubes A, B, C or D showed the presence of starch? Absence of starch? Explain the reason for this result.
- 2
- Which of the 4 test tubes A2, B2, C2 or D2 showed the presence of sugar? Absence of sugar? Explain the reason for this result.
- 3
- What was the reason for keeping the enzyme and starch solution at 37°C?

Conclusion

Label the diagram and explain which substances are digested in each organ.

The diagram shows a human figure from the waist up, with the digestive system highlighted. Labels 1 through 10 point to the following organs: 1. Mouth, 2. Esophagus, 3. Stomach, 4. Liver, 5. Gallbladder, 6. Duodenum (first part of small intestine), 7. Jejunum (middle part of small intestine), 8. Ileum (last part of small intestine), 9. Cecum (first part of large intestine), 10. Rectum.

Organ	Enzyme produced	Digestion

ASEXUAL REPRODUCTION IN PLANTS AND ANIMALS



Background

Timing: 2 × 45 minute lessons

To ensure the continuation of the species, plants and animals use two methods of **reproduction** – **sexual** and **asexual**. Sexual reproduction involves the fusing of two gametes, one from each partner, to form a new being. Asexual reproduction does not require a second partner. One or more offspring are produced from only one parent by the process of mitosis.

Vegetative reproduction is the propagation of plants using vegetative tissues such as stems, leaves and roots. This results in a plant that is genetically identical to the original plant. In some species, stems arch over and take root at their tips, forming new plants. For example, the horizontal above ground stems (**runners** or **stolons**) of the strawberry produce new daughter plants at alternate nodes. Underground stems such as **rhizomes**, **bulbs**, **corms** and **tubers** are used for asexual reproduction as well as for food storage.

Leaves of bryophyllum produce tiny plantlets along the leaf margin. Some plants like dandelion use their roots for asexual reproduction. Plants can be propagated asexually by taking stem cuttings and by grafting. Animals also reproduce asexually by various means.

Budding: The offspring develop as a growth on the body of the parent. In some species, such as jellyfish and many echinoderms, the buds break away and take up an independent existence. In corals, the buds remain attached to the parent and the process results in colonies of animals. Budding is also common among parasitic animals such as tapeworms.

Fragmentation and regeneration: As planaria and starfish grow to full size, they spontaneously break into eight or nine pieces. Each of these fragments develops into a mature organism. Bacteria and some protozoans reproduce by **binary fission** or **multiple fission**.

SAFETY RULES

- ⚠ Wear gloves, masks and lab coats while doing this experiment to avoid contamination.
- ⚠ Place the rooting hormone in a separate dish.
- ⚠ Clean the lab after the activity and pack up the material in an allocated place.
- ⚠ Clean your hands with disinfectant after planting.

Aim To grow plants by using stem cuttings, leaf cuttings, stolons, rhizomes, corms, tubers and bulbs.

Equipment

- geranium plant
- bryophyllum leaf
- strawberry plant
- ginger rhizome
- onion bulb
- gladiolus corm
- potato tuber
- a pair of secateurs
- sandy potting mixture
- Auxin (rooting hormone)
- 7 small plastic pots
- wooden sticks
- a clear plastic sheet
- a spray bottle
- empty egg container

Part A – Vegetative reproduction in plants

Method

Stem cutting: Geranium, fuchsias and impatiens can be grown by taking tip stem cuttings.

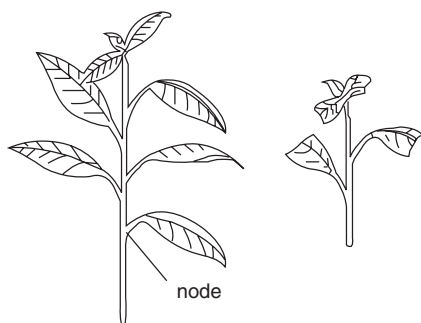


Figure 1: Stem cutting



Figure 2: Mini greenhouse for stem cutting

- 1 Use a pair of secateurs to take 2 cuttings from a healthy, disease-free geranium plant measuring 6 cm in length. The cutting must have at least five pairs of nodes. (A node is a place where the leaf joins the stem.)
- 2 Remove flowers, flower buds and leaves below the third node from the top of the stem.
- 3 Dip the cut end in auxin hormone to promote root growth.
- 4 Insert the cuttings one-third of their length into the potting medium. Space the cuttings just far enough apart to allow all leaves to receive sunlight.
- 5 Water after inserting the cuttings. Cover the cuttings with clear plastic sheet and place in indirect light. Avoid direct sun. Keep the medium moist until the cuttings have rooted. Mist the cuttings regularly with a spray bottle.
- 6 After a week observe whether the cuttings have produced any roots.

Leaf cuttings

- 7 Remove the tiny plantlets from a bryophyllum leaf and plant them in a pot.
- 8 Cover the cuttings with clear plastic sheet and place the pot in a shady area and spray with water regularly.
- 9 After a week, observe the tiny plantlet.

Runners or Stolons: Prostrate aerial stems are called runners or stolons. Adventitious roots develop from the runners near the ground surface. Strawberries, numerous grasses, some ferns and spiker plants can be grown from runners.

- 10 Bend the stem of a strawberry plant so that part of it is touching the ground. Cover the stem with soil and put a stone on top of the soil to keep the stem underground.
- 11 Keep the pot in a shady area and spray with water regularly.
- 12 After a week, observe if roots have developed from the runner.

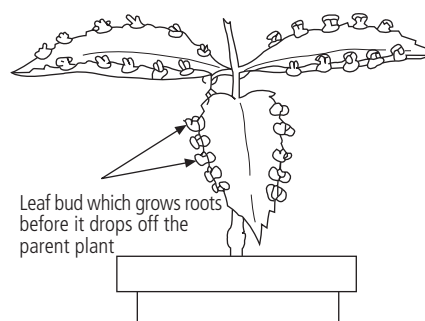


Figure 3: Bryophyllum leaf

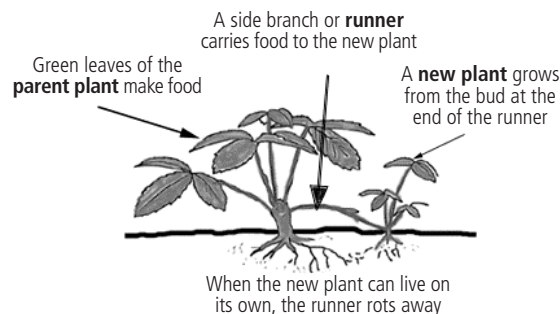


Figure 4: Strawberry runner

Rhizome, tuber, corm and bulb

- A **rhizome** is a modified, horizontal, underground stem of a plant that sends out roots and shoots from its nodes. Irises, ginger and some ferns produce rhizomes underground.
 - A **tuber** is a thickened part of the underground stem that stores starch. Potato, begonia and Jerusalem artichoke are tubers.
 - A **corm** is a short, thickened, vertical underground stem that stores starch. Gladiolus and crocus are corms.
 - **Bulbs** are specialised buds with thickened, fleshy scales. Onions, garlic, daffodils and hyacinths produce daughter bulbs.
- 13 **Rhizome:** Take a small piece of ginger rhizome and place it 1 cm below the soil. Cover it with soil and water. Place in a shady area and water it regularly. Observe it after a week.
 - 14 **Bulbs:** Take an onion bulb and plant it 6–8 cms deep. Cover it with the soil. Water it regularly and observe after a week.
 - 15 **Tubers:** Potatoes are grown from 'seed' potatoes that need to 'sprout' before planting. Place the seed potatoes in an egg box in a light airy position away from direct sunlight.

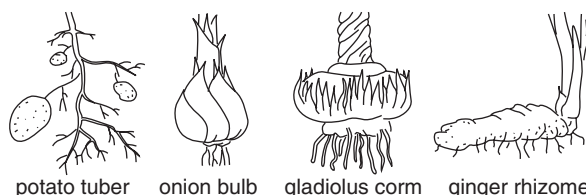


Figure 5: Tuber, bulb, corm and rhizome

- 16 The potato seed should be positioned so that the sprouts are uppermost. As the potatoes produce sprouts, remove all but the topmost sprouts to ensure that they receive all the goodness of the seed potato. The ideal sprout length at planting time is 2.5 cm. The sprouts should be green in colour. Insufficient light will cause white sprouts.
- 17 Dig a hole around 6 cm deep and place the potatoes in it with the sprouts pointing up. Cover the potatoes with soil, trying to avoid damaging any sprouts. Water the pot and place it in a shady area. Observe after a week.
- 18 Take a gladiolus corm and plant it 6 cm deep. Cover it with soil, water it regularly and observe it after one week.

Note: You can plant rhizomes, tubers, corms and bulbs in the same pot but at different depths as instructed. If possible, place the plants in a shade house or a glass house.

Results

- 1 On a separate sheet of paper, draw diagrams of the stem cutting, leaf cutting, runner, bulb, corm, tuber and rhizome after they have grown roots.

Analysis of results

- 1 What is meant by vegetative reproduction?
- 2 Which plants can be grown from stem cuttings and leaf cuttings?
- 3 What are the differences between rhizomes, corms, tubers and bulbs?
- 4 How are potatoes grown?
- 5 How can you grow a new plant from a strawberry runner?

Part B – Asexual reproduction in animals

Aim To observe slides and charts to understand how some animals reproduce asexually.

Equipment

- slides of hydra
- planaria
- amoeba
- sea anemone
- starfish undergoing asexual reproduction
- charts showing various types of asexual reproduction
- light microscope
- microscope lamp

Method

- 1 **Binary fission and multiple fission in amoeba:** Amoeba splits into 2 by binary fission during favourable conditions. During unfavourable conditions they form spores by multiple fission. Observe the chart and slide of amoeba dividing by binary fission.
- 2 **Budding in hydra:** Although capable of sexual reproduction, hydras reproduce most of the year by producing external asexual buds that eventually pinch off at the base and become new individuals. Observe the chart and slide of budding in hydra.

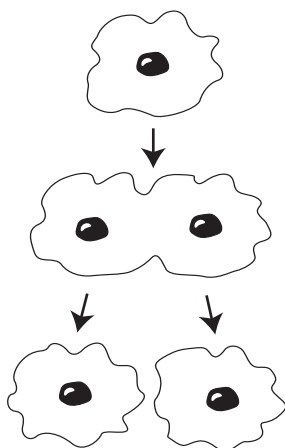


Figure 6: Binary fission by bacteria

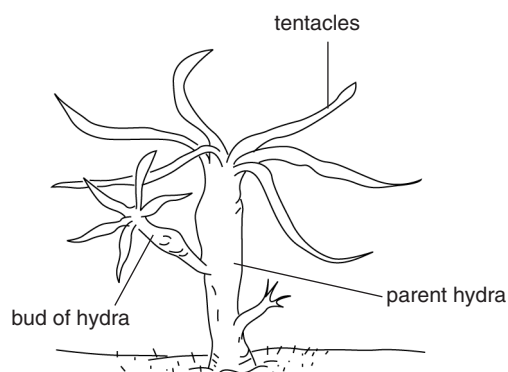


Figure 7: Budding by hydra

- 3 **Fragmentation and regeneration in planaria:** Under natural conditions, the planarian ordinarily fragments into two parts. The anterior part then regenerates a new tail while the posterior fragment develops a new head. Observe the chart and slide and of fragmentation in planarian.
- 4 **Fragmentation and regeneration in sea anemone and starfish:** A sea anemone splits along its length and develops into two individuals after necessary growth. In starfish, a cut arm regenerates all the missing parts to become a whole starfish. Observe the chart and slide of sea anemone and starfish regeneration.

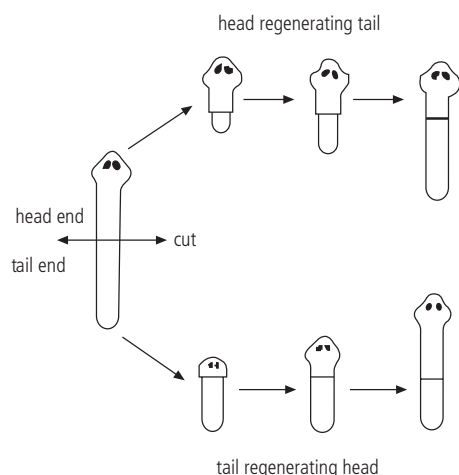


Figure 8: Fragmentation and regeneration in planaria

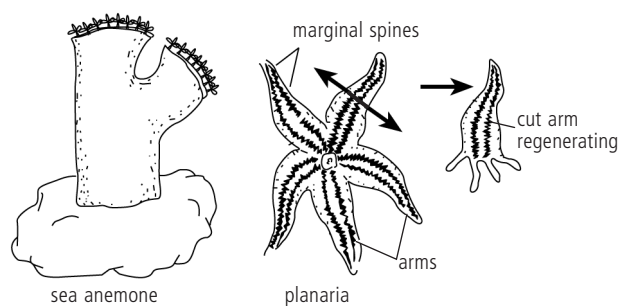


Figure 9: Fragmentation and regeneration in sea anemone and starfish

Results

- 1 On a separate sheet of paper, draw diagrams of the binary fission in amoeba, budding in hydra, regeneration in planaria and fragmentation and regeneration in sea anemone and starfish.

Analysis of results

- 1 When do amoebae reproduce by binary fission and by multiple fission?
- 2 How does hydra reproduce asexually?
- 3 How does a planarian reproduce by fragmentation and regeneration?
- 4 How do sea anemone and starfish reproduce asexually?

Conclusion

- 1 How do plants and animals reproduce asexually?
- 2 What are the advantages and disadvantages of different types of reproduction?

FOOD CHAINS, FOOD WEBS AND FOOD PYRAMIDS



Background

Timing: 2 × 45 minute lessons

Ecology is the study of the interactions of organisms with other organisms and with their physical environment. **Food chains and webs** are often used to show these interactions. Plants are known as **producers** because they use sunlight and inorganic materials to produce organic compounds. This occurs through the process of photosynthesis, where light energy is converted into chemical energy. Animals that feed upon plants are called **primary consumers**, while animals that eat other animals are known as **secondary** or **tertiary** consumers. **Scavengers** feed on dead organisms. **Decomposers** and **detrivores** break down non-living organic matter into materials that are then available to enter the food chain as nutrients.

A **food chain** shows how each living organism gets its food. A **food web** consists of many food chains which are interrelated and divided into separate feeding groups or **trophic levels**. The trophic level of an organism is its position in the energy flow pyramid of an ecosystem. A **food pyramid** helps us to visualise the fact that in an ecological system there need to be many producing organisms at the bottom level of the pyramid in order to sustain fewer organisms at the top level. Energy is transferred from one trophic level to the next trophic level in a food chain and food web. When one organism consumes another, about 5–10% of the chemical energy is transferred to the predator. 90% of energy is lost as heat energy to the environment. More pesticides and insecticides accumulate in organisms at higher trophic levels. This phenomenon is known as **biological magnification**.

Aim

- To group different organisms into their correct trophic levels.
- To construct food chains and food webs by using organisms given in a table.
- To analyse how energy and pesticides are transferred from one trophic level to the next higher trophic level.

Equipment

- A4 paper
- tables of names of organisms, reference books and charts

Part A – Food chain

A **food chain** shows how each living organism gets its food. A simple food chain starts with a producer. The producer is eaten by a primary consumer, which is eaten by the secondary consumer. A tertiary consumer eats the secondary consumer. Most food chains have no more than four or five links so that the animals at the end of the chain can get sufficient energy.

Phytoplankton are small microscopic producers found on the top layer of the ocean. They trap sunlight by photosynthesis. **Zooplankton** are organisms that feed on phytoplankton. Some zooplankton are herbivorous and others are carnivorous, feeding on smaller organisms.

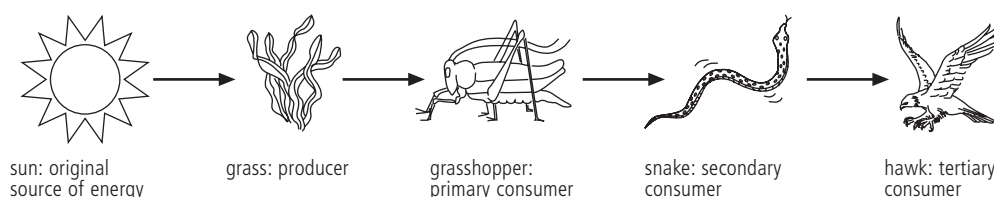


Figure 1: A terrestrial food chain

Method

- 1 Draw an aquatic food chain consisting of phytoplankton, zooplankton, fish and shark.

Analysis of results

- 1 Compare the terrestrial food chain with the aquatic food chain in the following table.

Trophic levels	Terrestrial food chain	Aquatic food chain
Producer		
Primary consumer		
Secondary consumer		
Tertiary consumer		

2 Where is the original source of energy coming from for both food chains?

Part B – Food webs

Food webs are more complex than food chains. In the food web shown below, many food chains are interconnected.

Note: the arrows are drawn from food source to consumers.

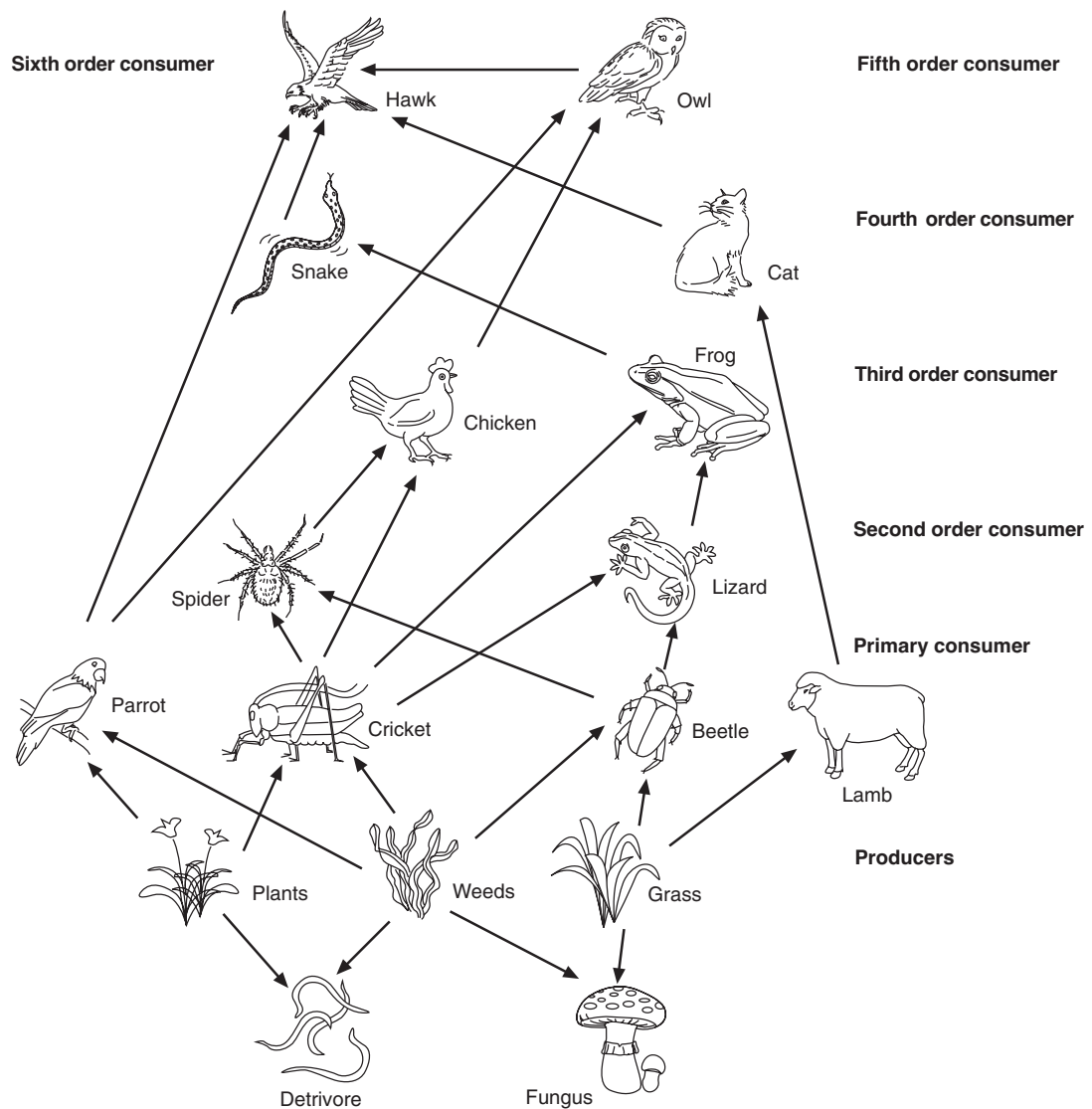


Figure 2: Food web of a terrestrial community

Construction of a marine food web

Method

- 1 Use the organisms in the table below to construct a marine food web. (Hint: Put the phytoplankton at the bottom of the page and the consumers of different trophic levels above the producers as shown in the terrestrial food web.)

Organisms in a marine food web			
Organism	Trophic type	Prey/food	Predators/grazers
Phytoplankton	Producers	photosynthesise	Krill and herbivorous zooplankton
Herbivorous zooplankton	Primary consumer	Phytoplankton	Krill and fish
Carnivorous zoo plankton	Secondary consumer	Krill	Fish, blue whales and birds
Krill	Primary consumer	Phytoplankton	Fish, blue whales, birds, penguins, leopard seal, other seals and baleen whale
Squid	Tertiary consumer	Krill, fish and herbivorous zooplankton	Fish, penguin, elephant seal, perm whale and small toothed whale
Fish	Tertiary consumer	Krill and squid	Penguin, leopard seal and elephant seal
Other birds	Tertiary consumer	Krill and fish	Leopard seal
Other seals	Tertiary consumer	Krill and carnivorous zooplankton	Small toothed whale
Elephant seal	Tertiary consumer	Fish and squid	Small toothed whale
Leopard seal	4th order consumer	Krill, fish, other birds and penguins	Small toothed whale
Penguin	4th order consumer	Krill, fish and squid	Leopard seal and small toothed whale
Baleen whale	Secondary consumer	Krill	Small toothed whale
Small toothed whale	5th order consumer	Other seal, leopard seal, elephant seal and baleen whale	None
Sperm whale	Tertiary consumer	Squid	None

Results

Draw a marine food web on a separate sheet of paper and submit it to your teacher.

Analysis of results

- 1 Compare the organisms in the terrestrial food web with the organisms in the marine food web you drew.

Trophic levels	Terrestrial food web	Marine food web
Producers		
Primary consumers		
Secondary consumers		
Tertiary consumers		
4th order consumers		
5th order consumers		
6th order consumers		
Detrivore		
Decomposer		

- 2 Would there be more predators or prey in these communities?
- 3 Where would humans fit in these food webs?
- 4 Describe how humans might change (or are currently changing) these food webs.
- 5 Explain what may happen to other organisms if the pollutants in the water decrease the phytoplankton. Which organisms would increase in population? Why? Which organisms would decrease in population? Why?
- 6 What would happen to other organisms in the web if small-toothed whales became extinct?
- 7 What might happen if phytoplankton was removed from the food web?



Part C – Food pyramids

A **food pyramid** helps us to visualise the fact that in an ecological system there should be many producing organisms at the bottom of the pyramid in order to sustain fewer organisms at the top level. High trophic levels contain fewer organisms than lower levels.

Energy in a food pyramid: It is estimated that in going from one trophic level to the next, about 90% of the energy is lost as heat energy. This means that in moving to the second trophic level, only 10% of the original energy is available. By the third trophic level, only 1% of the energy is available.

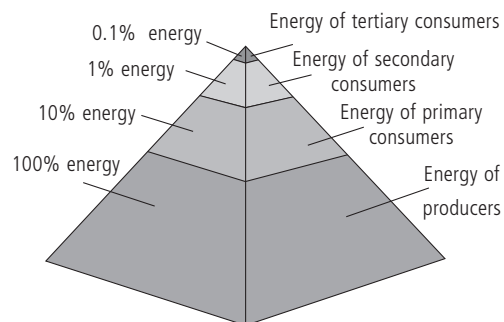


Figure 3: Energy transfer in a food pyramid

Method

- 1 Draw a diagram of a food pyramid consisting of grass and plants, cricket, birds and hawks.

Results

Submit your diagram to your teacher.

Analysis of results

Refer to the food pyramid in figure 3 to help you answer the questions

- 1 How does the flow of energy change as it moves from producers to the primary consumers, secondary consumers and then tertiary consumers?
- 2 Which organisms get the most energy? The least energy?
- 3 What will happen if the number of hawks increases?

Part D – Bioaccumulation and biomagnification

Bioaccumulation is the increase in concentration of a pollutant from the environment to the first organism in a food chain.

Biological magnification (or biomagnification) is the buildup of certain substances, such as DDT, in the bodies of organisms at higher trophic levels of food webs. DDT is a **non-biodegradable** substance that builds up in the fatty tissues of organisms. Organisms at lower trophic levels accumulate small amounts. Organisms at the next higher level eat many of these lower-level organisms and hence accumulate larger amounts. At the highest trophic levels, the increased concentrations in tissues may become toxic. In birds, for instance, DDT interferes with the deposition of calcium in the shells of the bird's eggs. The eggs laid are very soft and easily broken; birds so afflicted are rarely able to raise young and this causes a decline in their numbers.

Method

- 1 Draw a food pyramid consisting of algae, mayfly nymph, sunfish and great blue heron. Use dots to show how the DDT accumulates at each trophic level.

Results

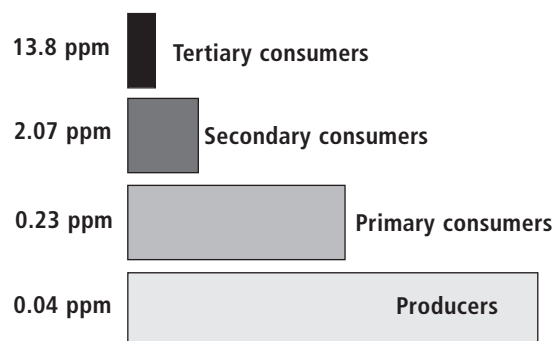
Submit your diagram to your teacher.

Analysis of results

- 1 How much DDT is found in producers? Primary producers? Secondary consumers? Tertiary consumers?
- 2 Which organism will have the highest amount of DDT in its tissue?
- 3 How does this accumulation of DDT affect the survival of the organisms?

Conclusion

- 1 How are food webs different from food chains?
- 2 Explain how energy is transferred from producers to consumers at different trophic levels. (Refer to the food pyramid.)
- 3 Define biomagnification and explain how DDT accumulates in different trophic levels.



The numbers are representative values of the concentration in the tissues of DDT and its derivatives

Figure 4: Biomagnification of DDT in a food pyramid

Timing: 2 × 45 minute lessons preparation, a day trip to the area of study, 2 × 45 minute lessons for follow-up.

Background

Scientists use two methods to survey the plants in a community, the **transect** method and the **quadrat** method. A **quadrat** consists of a square, rectangular or circular frame which is used to identify and calculate the abundance of plant species within a given area. Rectangular quadrats with a width-to-length ratio of 1:2, 1:4 or 1:8 are commonly used.

Transects are lines or belts that are set out in an area to study the type and abundance of a species. A **line transect** is made by laying out a tape measure or a rope marked at regular intervals along the area to be studied. The name and position of plant species are recorded and **profile diagrams** are drawn to scale. **Belt transect** is made by laying out two parallel lines to define the area to be studied.

SAFETY RULES

- ⚠ Do not remove organisms from the area of study, except for a small stem cutting for identification.
- ⚠ Place stem cuttings of a plants for identification in plastic bags and label them.
- ⚠ Do not trample on plants.

Aim To survey, identify and calculate the abundance and density of plants, and record the biotic and abiotic factors in a given area, using transect and quadrat methods.

School grounds activity

Before the field trip, explore the school grounds by practising both transect and quadrat methods. Test a soil sample and determine the height of a tree by using the methods given below. Use the field guide to identify plants.

Part A – Soil testing

- Equipment**
- soil sample
 - balance
 - distilled water
 - glass rod
 - 100 mL beaker
 - petri dish
 - pH meter with probe
 - pH paper.
 - 100 mL measuring cylinder

Method

- 1 Weigh 20 g of dry soil in a 100 mL beaker.
- 2 Add 20 mL of distilled water and stir with a glass rod.
- 3 Measure the pH of the soil-water mixture using pH paper or a pH meter. The table at right shows the pH scale that indicates whether the soil is acidic, neutral or alkaline.
- 4 Record your results in Table 1.

pH	Classification
>8.5	strongly alkaline
7.9–8.5	moderately alkaline
7.3–7.9	slightly alkaline
6.7–7.3	neutral
6.2–6.7	slightly acid
5.6–6.2	moderately acid
3.0–5.6	strongly acid

Results

Table 1

Indicators used to test the soil	Ph of the soil	Type of soil (acidic, alkaline or neutral)
pH paper.		
pH meter.		

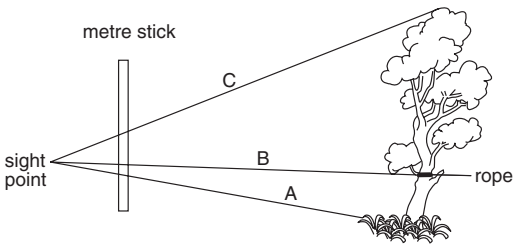
Part B – Measuring the height of a tree

Equipment

- metre ruler
- rope

Method

- 1 Measure 2 metres from the ground and tie a rope around the tree trunk at this height.
- 2 Move away from the tree to a distance that allows the entire tree, from the base to the top, to fit within the meter stick when held out in front of you.
- 3 Count the number of centimetres the tree covers (from A to C on the diagram) and the number of centimetres from the ground to the rope (from A to B on the diagram). Record these lengths in Table 2 and then calculate the actual height.



Results

Table 2

Tree height (cm) A to C	Rope height (cm) A to B	Actual height of the tree = Tree height ÷ rope height × 2 (m)

Part C – Transect method

Transects are lines or belts set out through an area for recording changes in the abundance of a plant species. **Line transect** is made by putting a tape measure across the area to be sampled. **Belt transect** is similar to line transect but the vegetation between two parallel lines is recorded. It may be considered as a widening of the line transect to form a continuous belt, or series of quadrats. The plants and/or animals inside the quadrat are then identified and their abundance estimated. The percentage cover of plant species can also be estimated. **Cover** is the area of the quadrat occupied by the above-ground parts of a species when viewed from above.

- Equipment**
- Soil-testing kit
 - plant and animal identification field guides
 - long tape measure
 - 1 m² frame (made from pvc pipe or wire)
 - labels
 - 2 plastic bags
 - scissors
 - 5 m of strong rope, pre-measured and marked at 1-m intervals
 - 10 wooden stakes for anchoring the transect
 - hammer
 - compass
 - thermometer
 - camera

Method



Figure 1: Line transect

- 1 Work in groups of 3–4 students. Each group must choose a different area for the transect.
- 2 Stretch 5 metres of string, pre-measured and marked at 1-metre intervals, across the area to be surveyed.
- 3 At each metre mark (1 m, 2 m, 3 m, 4 m and 5 m), record a description of any plants below or touching the string.
- 4 To calculate the frequency, count the number of each type of plant that occurs in each of these intervals. If there is no vegetation present, record that.
- 5 Move along the transect line and for each plant that touches, overlies or underlies the transect rope, record the name and the distance along the line that is intercepted by the plant.
- 6 For tall herbs, shrubs and trees, the distance intercepted must be determined by measuring the downward projection of the crowns onto the strip. Record all data as they are obtained. The use of symbols for species name will speed up your work.
- 7 When taking samples, be sure you get enough stem to show how leaves are arranged and any flower structures or fruit present.
- 8 Use the field guide to identify the plants as required.
- 9 After you return to the classroom, calculate the frequency, abundance, density, relative density and cover. Present your results to the other students.

Part D – Quadrat method

Quadrats may be located randomly or at regular intervals along a transect to calculate the abundance, density and cover of a plant species.

Method

- 1 Consider the height and the density of the plants in the study area when you are deciding what size quadrat to use. The quadrat must be large enough to contain a significant number of plants yet small enough to permit you to identify, count and measure the plants in a reasonable length of time. In general, the following quadrat areas are satisfactory:
 - herbs (including grasses) and tree seedlings: $1\text{ m} \times 1\text{ m} = 1\text{ m}^2$
 - shrubs and saplings: $4\text{ m} \times 4\text{ m} = 16\text{ m}^2$
 - trees: $10\text{ m} \times 10\text{ m} = 100\text{ m}^2$
- 2 If you are sampling mainly herbs and seedlings, mark off a 1 m^2 area with 4 m of string and stakes.
- 3 Count and identify each type of plant that falls inside the square area.
- 4 Repeat this procedure 4 times for an accurate sampling of the area.
- 5 If you are measuring a plant that is circular and hugs the ground, measure its diameter and then use $\text{Area} = \pi \times \text{diameter}$ to determine the area that it covers.
- 6 If the plant is a tall herb or shrub, measure the downward projection of the crown on the ground. Again, use the diameter to calculate the area.
- 7 For trees, determine cover by using the downward projection of the crown.
- 8 Measure the temperature of the air with a thermometer.
- 9 Record the cloud cover, sunlight intensity and wind velocity.
- 10 Take a sample of the soil in a plastic bag to analyse later in class.
- 11 Measure the height of trees by using the method given in part B.
- 12 Draw profile diagrams of the transect and quadrat and take photos.
- 13 Record your results in Table 3 and use the data to calculate the following:
 - **Abundance of a species** compares the number of plants of that species with the total number of plants of all species in the study area:
 $(\text{Number of plants of a certain species} \div \text{total number of plants}) \times 100\%$
 - **Density** is defined as the number of plants of a certain species per unit area:
 $(\text{Number of plants of a certain species} \div \text{total area sampled}) \times 100$
 - **Cover** is the percentage area of the quadrat covered by a particular plant species. To record percentage cover of species in a quadrat, look down on the quadrat from above and estimate the percentage cover occupied by each species (e.g. species A–D in figure 3). Species often overlap and there may be several different vertical layers. Therefore, percentage cover may add up to well over 100% for an individual quadrat. Estimation can be improved by dividing the quadrat into a grid of 100 squares each representing 1% cover. This can be done mentally by imagining 10 longitudinal and 10 horizontal lines of equal size superimposed on the quadrat, or physically by actually dividing the quadrat with string or wire attached to the frame at standard intervals.

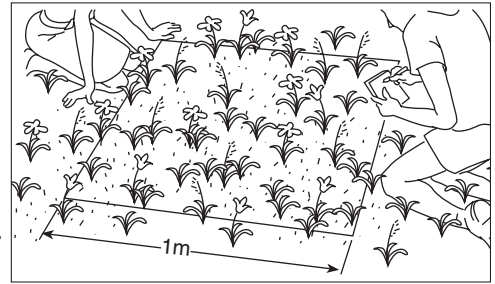


Figure 2: A quadrat

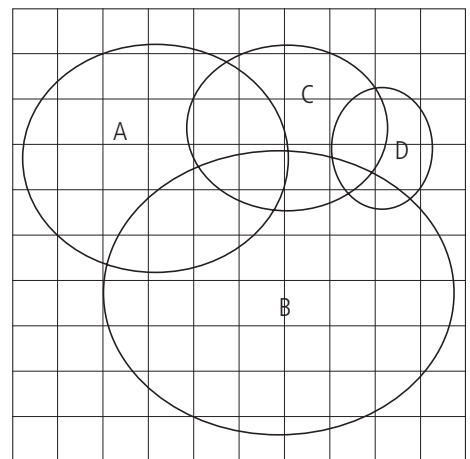


Figure 3: Percentage cover of different species of plants in a quadrat.

Results

Table 3 Measurement of abiotic factors

Abiotic factors	Transect	Quadrat
Temperature in °C		
Sunlight (cloudy or bright)		
Wind velocity (windy or still)		
Rainy day or clear day		
Soil type (sandy or clay)		
pH of the soil (acidic, alkaline or neutral)		

Table 4 Measurement of biotic factors in the quadrat

Number of a plant species (a)	Total number of plants (b)	Abundance of species ($a \div b \times 100$)	Density ($a \div 1 \text{ m}^2 \times 100$)

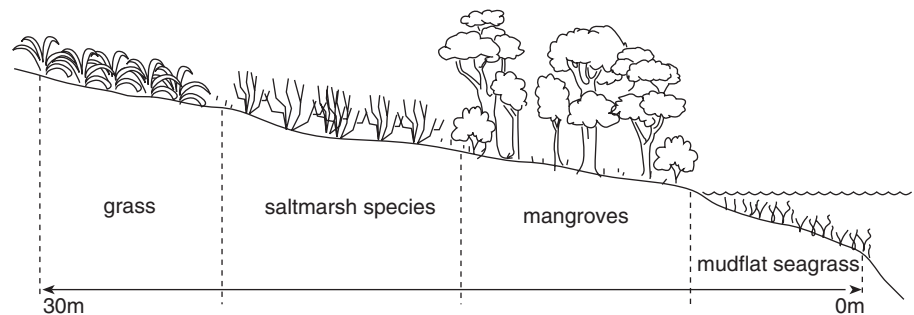


Figure 4: Profile diagram of a salt marsh

Profile diagram: A profile diagram is a scale drawing of the profile of plants along a transect line. It also shows the shape of the land surface and soil type. Draw profile diagrams of your transect and quadrat as shown.

Analysis of results

- Compile class results and answer the following questions.
- 1 Which plant species were most abundant? Which plant species were least abundant?
 - 2 What type of soil did you find in the area studied?
 - 3 What was the temperature on the day of your sampling?
 - 4 What variations did you notice at different intervals of the transect line?
 - 5 Did you notice any living organisms or traces of their presence in the area of study?

Conclusion

Describe the area you studied with the transect and quadrat methods.

SUGGESTED ANSWERS



Classification of animals

Part A Results

Table 1 Characteristics of vertebrates

Structures observed	Shark	Goldfish	Frog	Salamander	Snake
Backbone (present or absent)	Present	Present	Present	Present	Present
Skeleton (internal or external)	Internal	Internal	Internal	Internal	Internal
Symmetry (radial or bilateral)	Bilateral	Bilateral	Bilateral	Bilateral	Bilateral
Body covering (skin, fur, hair, feathers or scales)	Rough skin	Scales	Moist skin	Moist skin	Dry scales
How they move (legs, wings or fins)	Swims with fins	Swims with fins	Swims and hops with 4 legs, webbed feet	Swims and walks with 4 legs, webbed feet	Slides, has no legs
How they breathe (gills, lungs or skin)	Gills	Gills	Lungs and skin	Lungs and skin	Lungs
Ectothermic or endothermic	Ectothermic	Ectothermic	Ectothermic	Ectothermic	Ectothermic
Class	Fish	Fish	Amphibian	Amphibian	Reptile

Structures observed	Turtle	Duck	Rat	Bat
Backbone (present or absent)	Present	Present	Present	Present
Skeleton (internal or external)	Internal	Internal	Internal	Internal
Symmetry (radial or bilateral)	Bilateral	Bilateral	Bilateral	Bilateral
Body covering (skin, fur, hair, feathers or scales)	External shell	Feathers	Hair	Hair
How they move (legs, wings or fins)	Walks with 4 legs	Swims, flies and walks with 2 legs	Walks with 4 legs	Flies with wing-like structures and has 2 legs
How they breathe (gills, lungs or skin)	Lungs	Lungs	Lungs	Lungs
Ectothermic or endothermic	Ectothermic	Ectothermic	Ectothermic	Ectothermic
Class	Reptile	Bird	Mammal	Mammal

Analysis of results

1 Vertebrates have backbones. Invertebrates do not have backbones. **2** Structures of vertebrates. **Fins:** fish. **Gills:** fish. **Scales:** fish and reptiles. **Feathers:** birds. **Legs:** amphibians, some reptiles, birds, mammals. **Lungs:** amphibians, reptiles, birds, mammals. **Hair:** mammals. **Wings:** birds and bats. **3** Characteristics of vertebrates. **Fish:** Body covered with scales; have fins; swim in water; breathe with gills. **Amphibians:** Body covered with moist skin; have 4 webbed feet; swim in water and hop on land; breathe with lungs and skin. **Reptiles:** Body covered with dry scales; breathe with lungs; most have legs, snakes have no legs. **Birds:** Body covered with feathers; have wings and most can fly; have two legs and breathe with lungs. **Mammals:** Body covered with hair or fur, breathe with lungs; have 2 or 4 legs and can walk, run and swim.

1 Classification of animals (cont'd)

Part B

Results

Table 2 Characteristics of invertebrates

Structures observed	Sponge	Jellyfish	Flatworm	Roundworm	Earthworm	Snail
Presence of back bone	No	No	No	No	No	No
Shell or spiny skin	None	None	None	None	None	Shell
Symmetry (radial, bilateral or asymmetrical)	Asymmetrical	Radial	Bilateral	Bilateral	Bilateral	Part spiral
Number of segments	None	None	None	None	Segmented	None
Shape of body	Irregular	Umbrella Shape	Flat	Round	Round	Flat
Number and type of legs	None	None	None	None	None	None
Wings	None	None	None	None	None	None
Tentacles or antennae	None	Tentacles	None	None	None	Antennae
Ectothermic or endothermic	Endothermic	Endothermic	Endothermic	Endothermic	Endothermic	Endothermic
Phylum	Porifera	Cnidaria	Platyhelminthes	Nematoda	Annelida	Mollusca

Structures observed	Millipede	Centipede	Butterfly	Crab	Starfish
Presence of back bone	No	No	No	No	No
Shell or spiny skin	None	None	None	Shell	Spiny skin
Symmetry (radial, bilateral or asymmetrical)	Bilateral	Bilateral	Bilateral	Bilateral	Radial
Number of segments	Segmented	Segmented	Segmented	Segmented	None
Shape of body	Round	Round	Has head, thorax and abdomen	No head and thorax	Star-shaped
Number and type of legs	Many pairs of legs	100 pairs of legs	6 legs	8 legs	Many legs and 5 arms
Wings	None	None	Has 2 pairs of wings	None	None
Tentacles or antennae	Antennae	Antennae	Antennae	Antennae	None
Ectothermic or endothermic	Endothermic	Endothermic	Endothermic	Endothermic	Endothermic
Phylum	Arthropoda	Arthropoda	Arthropoda	Arthropoda	Echinodermata

Analysis of results

1 Invertebrates do not have backbones. 2 Structures of invertebrates. **External shell:** snail and crab. **Asymmetrical:** sponges. **Bilateral symmetry:** flatworm, roundworm, earthworm, millipede, centipede, butterfly, crab. **Radial symmetry:** jellyfish, starfish. **Segmented:** earthworm, millipede, centipede, butterfly, crab. **Not segmented:** sponge, jellyfish, flatworm, roundworm, starfish. **Wings:** butterfly. **Many legs:** millipede, centipede, starfish. 3 Identification of organisms and phyla. **A tapeworm:** Non-segmented flat body, no legs, bilateral symmetry, Platyhelminthes. **B hydra:** Hollow body, tentacles, radial symmetry, Coelenterate. **C starfish:** 5 arms, spiny skin, non-segmented body, radial symmetry, Echinoderm. **D jellyfish:** Umbrella-shaped body, tentacles, radial symmetry, Coelenterate. **E earthworm:** Non-segmented flat body, no legs, bilateral symmetry, Platyhelminthes. **F snail:** Spiral shell, soft non-segmented body, no jointed feet, mollusc. **G ant:** 3-segmented body, 3 pairs of jointed feet, bilateral symmetry, Arthropod. **H millipede:** segmented body, 2 pairs of legs per segment, bilateral symmetry, Arthropod. **I spider:** 2-segmented body, 4 pairs of legs, bilateral symmetry, Arthropod. **J crayfish:** Segmented body, numerous jointed legs, external shell, bilateral symmetry, Arthropod. **K centipede:** Segmented body, 1 pair of legs per segment, bilateral symmetry, Arthropod.

Conclusion

1 Vertebrates have internal skeletons and backbones made up of vertebrae. Fish, amphibians and reptiles are endothermic, birds and mammals are ectothermic. All vertebrates have bilateral symmetry.

Invertebrates do not possess internal skeletons or backbones. Some have external shells. Some are asymmetrical, some are bilateral and others or radial in symmetry. All invertebrates are endothermic. Some possess legs, wings, antennae and eyes. Some have segmented bodies and others possess non-segmented, round or flat bodies.



Classification of plants

Structures observed	Moss	Liverwort	Lichen	Fern	Pine
Stem (herbaceous, woody or hollow or solid)	Small herbaceous	Not present	Not present	Solid herbaceous	Woody, solid
Roots (tap root or fibrous roots)	Root-like structures	Root-like structures	None	Fibrous roots	Tap root system
Leaves (long, narrow, flat or needle-shaped)	Small	Liver-shaped	Crusty	Long and branched	Needle-like
Veins (net-like or parallel)	None	None	None	None	Has veins
Vascular or non-vascular	Non-vascular	Non-vascular	Non-vascular	Non-vascular	Vascular
Flowers (5 or 6 petals)	None	None	None	None	None
Cones	None	None	None	None	Present
Spores	Present	Present	Present	Present	None
Seeds (1 or 2 cotyledon)	None	None	None	None	Present
Phylum or class	Bryophyta Musci	Bryophyta Hepaticea	Fungi + algae	Pteropsida Filicophyta	Gymnosperm Coniferae

Structures observed	Geranium	Lily	Bean	Corn
Stem (herbaceous woody or hollow)	Solid herbaceous	Hollow, herbaceous	Solid, herbaceous	Hollow, herbaceous
Roots (tap root or fibrous roots)	Tap root system	Fibrous roots	Tap root system	Fibrous roots
Leaves (long, narrow, flat or needle-shaped)	Flat and broad	Long and narrow	Flat and broad	Long and narrow
Veins (net-like or parallel)	Net-like	Parallel	Net-like	Parallel
Vascular or non-vascular	Vascular	Vascular	Vascular	Vascular
Flowers (5 or 6 petals)	5 petals	6 petals	5 petals	6 petals
Cones	None	None	None	None
Spores	None	None	None	None
Seeds (1 or 2 cotyledon)	None	None	2 cotyledons	1 cotyledon
Phylum or group	Angiosperm, Dicot	Angiosperm, Monocot	Angiosperm, Dicot	Angiosperm, Monocot

Analysis of results

1 Vascular plants possess a vascular system. Non-vascular plants do not have vascular systems. **2** Gymnosperms and angiosperms have vascular systems. **3** Bryophytes and filicophytes do not have vascular systems. **4** Lichen is difficult to classify because fungi and algae live in a symbiotic relationship. **5 a** Angiosperms have flowers and seeds within fruits. Gymnosperms do not flower and have seeds within cones. **b** Monocots have one cotyledon, hollow stems and long leaves with parallel veins. Dicots have woody or herbaceous stems and net-like veins. Dicot flowers have 4–5 petals and monocots have 3–6 petals. **c** Bryophytes have small, stem-like structures, root-like structures and small leaves. They have no vascular system. They produce spores. Filicophytes have large stems, fibrous roots, large leaves and spores. They have no vascular system.

Conclusion

1 Plants are classified by noting whether they have vascular systems or not, woody or hollow stems, reproduce by spores or seeds, produce flowers or not and the type of venation.



Construction of a dichotomous key for arthropods

Results

Dichotomous keys External shell: crab, Wings: butterfly, 4 pairs of legs: spider, 100 pairs of legs: centipede, More than 100 pairs of legs: millipede.

Analysis of results

crabs (Crustacea): External shell, 5 pairs of legs, antennae. **butterflies (Insects):** 3 pairs of legs, no shell, 2 pairs of wings, antennae. **spiders (Arachnida):** 3 pairs of legs, no wings, no shell. **millipedes (Diplopoda):** More than 100 pairs of legs, no shell, no wings. **centipedes (Chilopoda):** 100 pairs of legs, no wings and no shell.

3 Construction of a dichotomous key for arthropods (cont'd)

Conclusion

1 Dichotomous keys are constructed to identify known and unknown animals, plants and other organisms. **2** The same dichotomous key cannot be used if a bee, a shrimp and a scorpion are included. This is because they have different characteristics which are not included in the key. **3** If an arthropod has no wings, 3 pairs of legs and a body divided into three segments, it will be placed in Insecta.



4 Dissection of a sheep's pluck

Analysis of results

Lungs

1 The lungs are pale pink in colour and feel firm and spongy. The left lung is divided into 2 lobes: an upper lobe and a lower lobe. The right lung is divided into 3 lobes: an upper lobe, middle lobe and a lower lobe. When we breathe in, air enters through the nostrils and goes to the pharynx, larynx, trachea, bronchi and bronchioles and finally to small sacs called alveoli. **2** The cartilage rings around the trachea support the trachea and prevent it from collapsing. **3** The lung tissue floated in water because it is light and spongy and contained air-filled alveoli. **4** Oxygen diffuses from the alveoli to the blood and carbon dioxide diffuses out of the blood into the alveoli.

Heart

5 The heart is divided into 4 chambers called the right atrium, left atrium, right ventricle and left ventricle. **6** The right side of the heart is thinner. **7** The left ventricle is the thickest. **8** Ventricles have thicker walls because blood is pushed out of the heart by ventricles. **9** Bicuspid and tricuspid valves between the atria and the ventricles allow blood to flow only from the atria to the ventricles. **10** The aorta is the largest blood vessel. **11** There is no difference between the sheep's heart and the model human heart, except in size. **12** The semi lunar valves situated between the pulmonary artery and the right ventricle allow blood to flow from the right ventricle to the pulmonary artery.

13

Structures/blood vessels	Functions
Right atrium	Receives deoxygenated blood from the body.
Right ventricle	Pumps deoxygenated blood to the lungs.
Left atrium	Receives oxygenated blood from lungs.
Left ventricle	Pumps oxygenated blood to the body.
Gall bladder	Stores bile.
Inferior venacava	Brings deoxygenated blood from lower parts of the body.
Superior venacava	Brings deoxygenated blood from upper parts of the body.
Pulmonary arteries	Take deoxygenated blood from the right ventricle to the lungs.
Pulmonary veins	Bring oxygenated blood from the lungs to the left atrium
Cardiac arteries	Supply oxygen and nutrients to the heart muscles.
Aorta	Takes oxygenated blood from the left ventricle to different parts of the body.

Liver

14 The liver is reddish-brown in colour and is divided into 2 lobes. It is the largest organ in the body. **15** The gall bladder is greenish in colour and it stores bile.

Conclusion

When the sheep's organs were compared with the organs of the human torso model, no differences in structure could be seen. Only the size, texture and colour were different.



5 Dissection of a rat

Analysis of results

Chest cavity

1 Lungs, heart, trachea and oesophagus were seen in the chest cavity. **2** The diaphragm separates the chest cavity from the abdominal cavity.

Digestive system

3 A rat has 4 sharp incisors to gnaw at food and 12 molars for grinding food. Canines and premolars are absent. There is an empty space between the incisors and molars called a diastem. **4** The stomach is J-shaped and is muscular. **5** The liver is the largest organ and it occupies one-third of the abdominal cavity. **6** The small intestine is narrower in diameter and longer in length than the large intestine. **7** The caecum is situated at the junction of small intestine and large intestine. It contains microbes, which help in digesting cellulose.

Urino-genital system

Excretory system **8** Kidneys are reddish-brown, bean-shaped organs situated on either side of the backbone. **9** Ureters carry urine from the kidneys to the urinary bladder for storage. Urine is expelled via the urethra. **Reproductive system** **10** Functions of the reproductive organs: Penis: Expels sperm and urine. Testes: Produce and store sperm. Ovaries: Produce ova. Uterus: Foetuses develop in the uterus. **11** Rats have a V-shaped uterus with 2 branches, whereas humans have only one small, un-branched uterus. **12** A rat's uteri can contain many foetuses.

Conclusion

The main differences between the rat and human systems are:

- Rats have only incisors and molars. Humans have incisors, canines, premolars and molars.
- Rats have a large caecum but humans have a small appendix.

5 Dissection of a rat (cont'd)

- Rats have a longer small intestine compared to the human small intestine.
- Female rats have a V-shaped and branched uterus that holds many foetuses. The human uterus is un-branched and usually holds only one foetus.
- Rats have common urino-genital openings and humans have separate urinary and genital openings.



Digestive enzymes

Part A – The digestion of milk fats by pancreatic lipase

Results

Table 1

Test tube	Presence of fatty acids (yes/no)	Rate of digestion (fast, slow or no digestion)	Colour of the litmus after 1.5 hrs	Explanation of each result
1	Yes	Fast	Transparent red	Bile salts and lipase emulsified and digested the cream.
2	No	None	Milky Blue	No reaction due to absence of lipase
3	Yes	Slow	Transparent red	No reaction due to absence of bile salts
4	No	None	Milky Blue	No reaction due to absence of the cream

Analysis of results

1 Test tube 1 contained both bile salts and the enzyme lipase so the digestion of fat was faster. Test tube 2 did not contain the enzyme lipase so no reaction occurred. Test tube 3 did not contain bile salts, so the fat was digested slowly. Test tube 4 did not contain fat, so digestion could occur.

2 When fat is digested, fatty acids are formed which changed the colour of litmus to red.

Part B – Digestion of proteins by pepsin and trypsin

Results

Table 2

Test tube	Appearance of egg white				Explanation of results
	Before		After		
	Food size	Food appearance	Food size	Food appearance	
C-2	5 mm ³	White solid	5 mm ³	White solid	Did not contain pepsin. No digestion took place.
C-8	5 mm ³	White solid	5 mm ³	White solid	Did not contain trypsin. No digestion took place.
P-2	5 mm ³	White solid	smaller pieces	Broken	Contained pepsin at pH2. Digestion took place.
P-8	5 mm ³	White solid	5 mm ³	White solid	Contained pepsin at pH8. No digestion took place.
T-2	5 mm ³	White solid	5 mm ³	White solid	Contained trypsin at pH2. No digestion took place.
T-8	5 mm ³	White solid	smaller pieces	Broken	Contained trypsin at pH8. Digestion took place.

Analysis of results

1 The egg white turned dark pink when copper sulfate was added. This indicates the presence of protein. There was no colour change in test tube 2 because it contained only water. **2** When protease was added, the egg white was digested. In the control test tube no digestion took place. **3** At pH2, pepsin digested the egg white completely. **4** At pH8, trypsin digested egg white completely. **5 a** Trypsin digests protein in the small intestine because its optimum pH is 8. **b** Pepsin digests protein in the stomach because its optimum pH is 2. **c** The water baths were kept at 37°C because that is the optimum temperature for digestive enzymes to react in the human body.

6 Digestive enzymes (cont'd)

Part C – Digestion of carbohydrates by pancreatic amylase

Results

Table 3

Test tubes	pH	Starch test (colour)	Presence of starch (yes/no)	Test tubes	pH	Sugar test (colour)	Presence of maltose (yes/no)	Brief explanation of each result
A	7	Brown	No	A 2	7	Green	Yes	Starch is digested to maltose due to the presence of amylase.
B	7	Purple	Yes	B2	7	Blue	No	As there was no enzyme, starch is not digested.
C	7	Brown	No	C2	7	Blue	No	As starch is not present no digestion took place.
D	3	Brown	No	D2	3	Blue	No	As the pH was acidic starch is not digested.

Analysis of results

1 Test tube B showed the presence of starch. Test tubes A, C, and D had no starch. **2** Test tube A2 showed the presence of maltose. In test tubes B2, C2 and D2 maltose was absent. **3** Since test tube A contained starch and amylase, starch was decomposed to maltose. Test tube B had no enzyme, test tube C had no starch and test tube D had starch, acid and enzyme. Due to these factors, starch was not decomposed to maltose. **4** 37°C is the temperature of our body and the enzymes react best at this temperature.

Conclusion

1 Salivary glands, salivary amylase, starch is digested to maltose. **2** Oesophagus, none, conveys food to the stomach. **3** Liver, gall bladder, bile salts, emulsifies fats. **4** Stomach, pepsin, digests proteins. **5** Pancreas, amylase and lipase, digests starch and fats. **6** Small intestine, many enzymes, all the food is digested. **7** Large intestine, none, water is absorbed. **8** Anus, none, no digestion and waste is excreted.



Asexual reproduction in plants and animals

Part A

Analysis of results

1 Vegetative reproduction is the propagation of plants using vegetative tissues such as stems, leaves and roots. This results in a plant that is genetically identical to the original plant. **2** Geraniums, fuchsia and impatiens can be grown from stem cuttings and bryophyllum can be grown from leaf cuttings. **3** A rhizome is a modified, horizontal underground stem of a plant that sends out roots and shoots from its nodes. A tuber is a thickened part of the underground stem that stores starch. A corm is a short, thickened, vertical underground stem that stores starch. Bulbs are specialised buds with thickened, fleshy scales. **4** Potatoes can be grown from sprouts that grow from seed potatoes. **5** When a strawberry stem is bent and placed under the soil, new roots will grow from the stem.

Part B

Analysis of results

1 During favourable conditions, amoeba reproduce by binary fission. During unfavourable conditions, it reproduces by multiple fission and forms spores. **2** Hydra reproduces by forming asexual buds that eventually pinch off at the base and become new individuals. **3** The planarian fragments into two parts. The anterior part then regenerates a new tail while the posterior fragment develops a new head. **4** A sea anemone splits along its length and develops into two individuals after necessary growth. In starfish, a cut arm regenerates all the missing parts to become a whole starfish.

Conclusion

Asexual reproduction is fast and maintains the characteristics of the parent organisms. But if the conditions change suddenly, all the offspring will die. Sexual reproduction is slower but the offspring are not identical to the parents. They can withstand sudden changes in the environment due to the variability in their characteristics.



Food chains and food webs

Part A – Food chain

Results

1 Aquatic food chain Producer: Phytoplankton; Primary consumer: Zoo plankton; Secondary consumer: Fish; Tertiary consumer: Shark.

Analysis of results

Trophic levels	Terrestrial food chain	Aquatic food chain
Producer	grass	phytoplankton
Primary consumer	cricket	zoo plankton
Secondary consumer	snake	fish
Tertiary consumer	hawk	shark

2 The original source of energy is coming from the sun.

8 Food chains and food webs (cont'd)

Part B – Food webs

Analysis of results

Trophic levels	Terrestrial food web	Marine food web
Producers	grass, weeds, plants	phytoplankton
Primary consumers	cricket, beetle, parrot, lamb	herbivorous zoo plankton, krill
Secondary consumers	spider, lizard	carnivorous zoo plankton, baleen whale
Tertiary consumers	hawk	shark
Tertiary consumers	chicken, frog	squid, fish, other birds, sperm whale, other seals, elephant seal
Fourth order consumers	snake, cat	leopard seal, penguin
Fifth order consumers	owl	small toothed whale
Sixth order consumers	hawk	none
Detrivore	worms	none
Decomposer	fungus	none

2 More prey is found than predators in this community. **3** Humans would be above the eel. **4** Since humans are fishing indiscriminately, the number of secondary and tertiary consumers are decreasing. This in turn would decrease the 4th and 5th level consumers. **5** If pollutants kill the phytoplankton, the number of zoo plankton would decrease. This in turn will affect all the other consumers in the food web. **6** If small-toothed whales become extinct, the number of fourth order consumers will increase. As a consequence, the number of third order consumers will decrease. **7** If phytoplankton are removed, the number of zoo plankton and all the other consumers will decrease.

Part C – Food pyramids

Analysis of results

1 Producers get 100% energy. Primary consumers get 10% energy. Secondary consumers get 1% energy. Tertiary consumers get 0.1% energy.

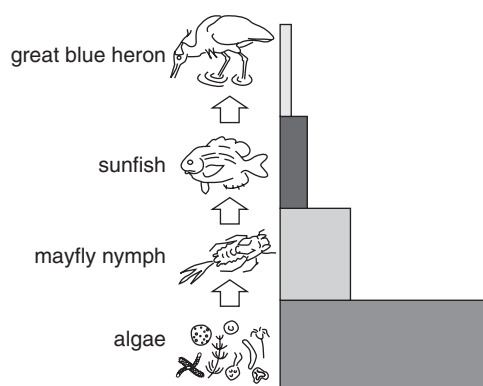
2 Producers get the most energy and tertiary consumers get the least energy.

Conclusion

1 Most animals are part of more than one food chain and eat more than one kind of food in order to meet their food and energy requirements. These interconnected food chains form a food web. Food webs are more complex than food chains. **2** Producers get 100% of the energy. But the consumers get less energy as you go up the trophic level. The tertiary consumers get the least energy, which is 0.1%. **3** If the number of hawks increases, the number of birds will decrease and this will in turn affect the number of insects. Insect numbers will increase and consequently the producer numbers will decrease.

Part D – Bioaccumulation and biomagnification

Results



Analysis of results

1 Producers 0.04 ppm; Primary producers 0.23 ppm, Secondary consumer: 2.07 ppm; Tertiary consumer: 13.8 ppm **2** Tertiary consumers such as the great heron and birds of prey will have the highest amount of DDT in their tissues. **3** DDT interferes with the deposition of calcium in the shells of the bird's eggs. The eggs laid are very soft and easily broken; birds so afflicted are rarely able to raise young and this causes a decline in their numbers.

Conclusion

1 Most animals are part of more than one food chain and eat more than one kind of food in order to meet their food and energy requirements. These interconnected food chains form a food web. Food webs are more complex than food chains. **2** Producers get 100% of the energy. The next trophic level gets only 10% of the original energy and 90% of the energy is lost as heat energy. By the third trophic level, only 1% of the energy is available. **3** Biological magnification is the build up of certain substances, such as DDT, in the bodies of organisms at higher trophic levels of food webs. Organisms at lower trophic levels accumulate small amounts. Organisms at the next higher level eat many of these lower level organisms and hence accumulate larger amounts. At the highest trophic levels the increased concentrations in tissues may become toxic.



9 Surveying an ecosystem

As this is a field trip, answers will vary depending on the area studied.

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